(Recap) Transformer

Wei Xu

(many slides from Greg Durrett)

Math Review

Dot Product

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\mathbf{a}^\top \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Bilinear Product

$$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}$$

$$\begin{aligned} \mathbf{a}^\top W \mathbf{b} = a_1(w_{11}b_1 + w_{12}b_2 + w_{13}b_3) \\ + a_2(w_{21}b_1 + w_{22}b_2 + w_{23}b_3) \\ + a_3(w_{31}b_1 + w_{32}b_2 + w_{33}b_3) \end{aligned}$$

Readings

- “The Annotated Transformer” by Sasha Rush
  https://nlp.seas.harvard.edu/2018/04/03/attention.html
- “The Illustrated Transformer” by Jay Lamar
  http://jalammar.github.io/illustrated-transformer/

- Jurafsky+Martin Chapter 7

Jurafsky+Martin NLP Textbook

Volume I: Large Language Models

Chapter

1: Introduction
2: Words and Tokens
3: N-gram Language Models
4: Logistic Regression and Text Classification
5: Embeddings
6: Neural Networks
7: Transformers and Pretraining
8: Post-training

Volume II: Advanced LLM Topics and Tools

Chapter

9: Masked Language Models
10: Interpretability
11: Information Retrieval and RAG
12: Agents [not written yet]
13: Machine Translation
14: RNNs and LSTMs
15: Phonetics and Speech Feature Extraction
16: Automatic Speech Recognition
17: Text-to-Speech

Volume III: Annotating Linguistic Structure

Chapter

18: Sequence Labeling for Parts of Speech and Named Entities
19: Context-Free Grammars and Constituency Parsing
20: Dependency Parsing
21: Information Extraction: Relations, Events, and Time
22: Semantic Role Labeling and Argument Structure
23: Lexicons for Sentiment, Affect, and Connotation
24: Coreference Resolution and Entity Linking
25: Discourse Coherence
26: Conversation and its Structure

Appendix (will be just on the web)

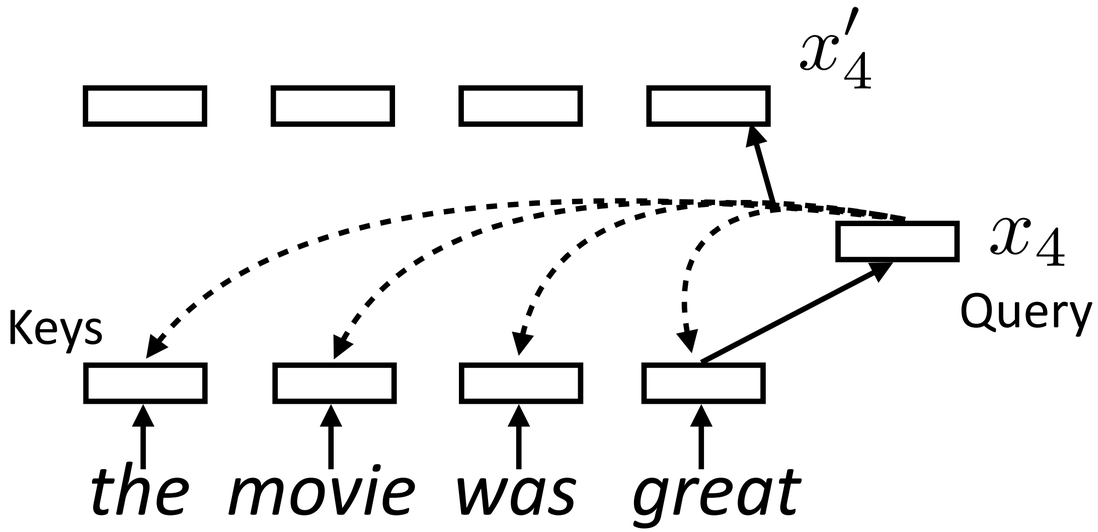
A: Hidden Markov Models
B: Naive Bayes Classification

Self-Attention

- Each word forms a “query” which then computes attention over each word

$$\alpha_{i,j} = \text{softmax}(x_i^\top x_j) \quad \text{scalar}$$

$$x'_i = \sum_{j=1}^{n} \alpha_{i,j} x_j \quad \text{vector = sum of scalar * vector}$$

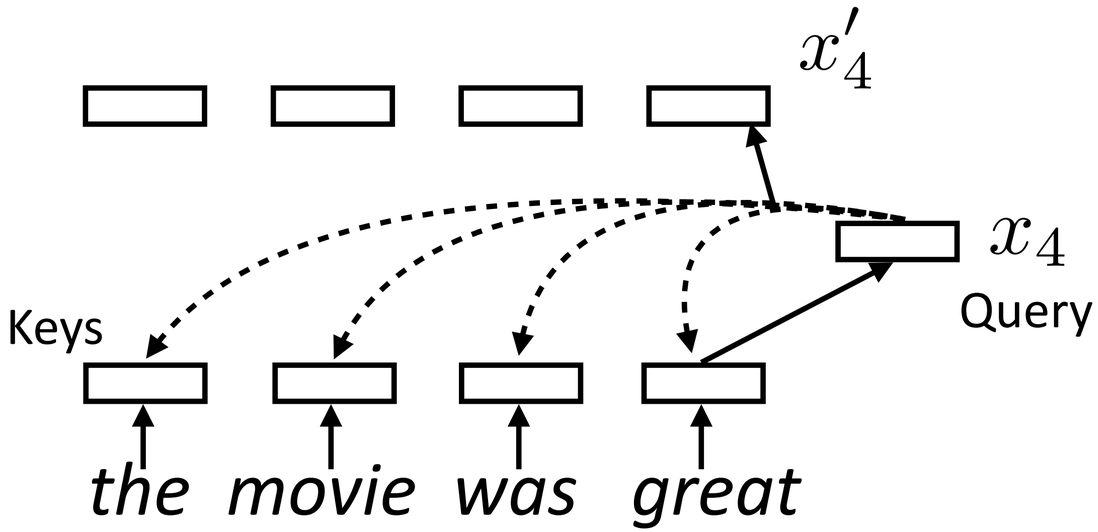


Self-Attention

- Each word forms a “query” which then computes attention over each word

$$\alpha_{i,j} = \text{softmax}(x_i^\top x_j) \quad \text{scalar}$$

$$x'_i = \sum_{j=1}^{n} \alpha_{i,j} x_j \quad \text{vector = sum of scalar * vector}$$



Keys

Queries

(tokens attending to other tokens)

	The	movie	was	great
The	0.25	**0.55**	0.13	0.07
movie	0.10	0.30	0.12	**0.48**
was	0.08	**0.42**	0.35	0.15
great	0.05	**0.60**	0.10	0.25

row-wise softmax

$$x'_{\text{great}} = \alpha_{41} x_{\text{the}} + \alpha_{42} x_{\text{movie}} + \alpha_{43} x_{\text{was}} + \alpha_{44} x_{\text{great}}$$

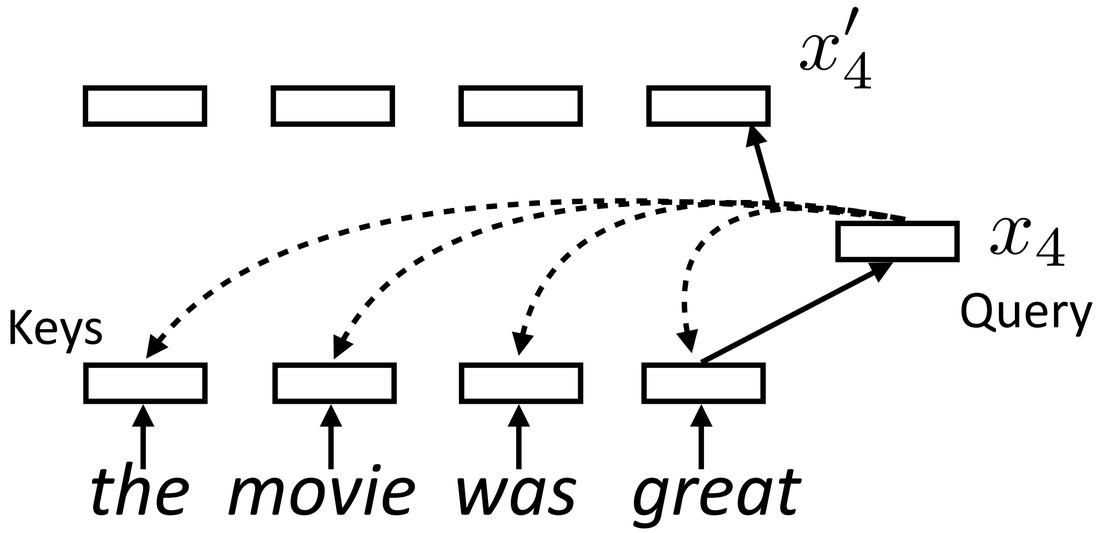
$$= 0.05 x_{\text{the}} + 0.60 x_{\text{movie}} + 0.10 x_{\text{was}} + 0.25 x_{\text{great}}$$

Self-Attention

- Each word forms a “query” which then computes attention over each word

$$\alpha_{i,j} = \text{softmax}(x_i^\top x_j) \quad \text{scalar}$$

$$x'_i = \sum_{j=1}^{n} \alpha_{i,j} x_j \quad \text{vector = sum of scalar * vector}$$



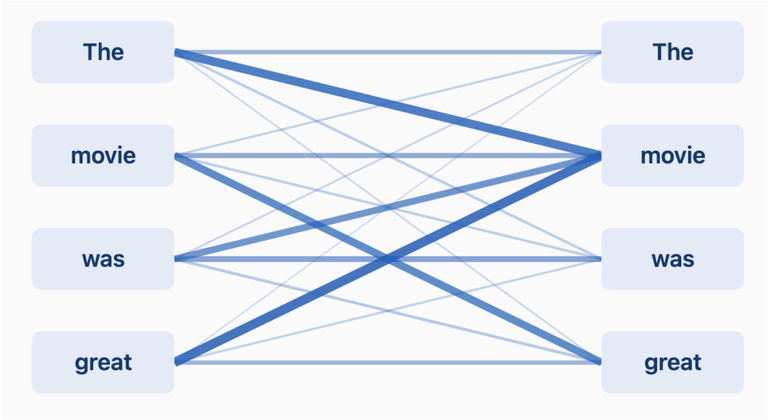
Keys

Queries

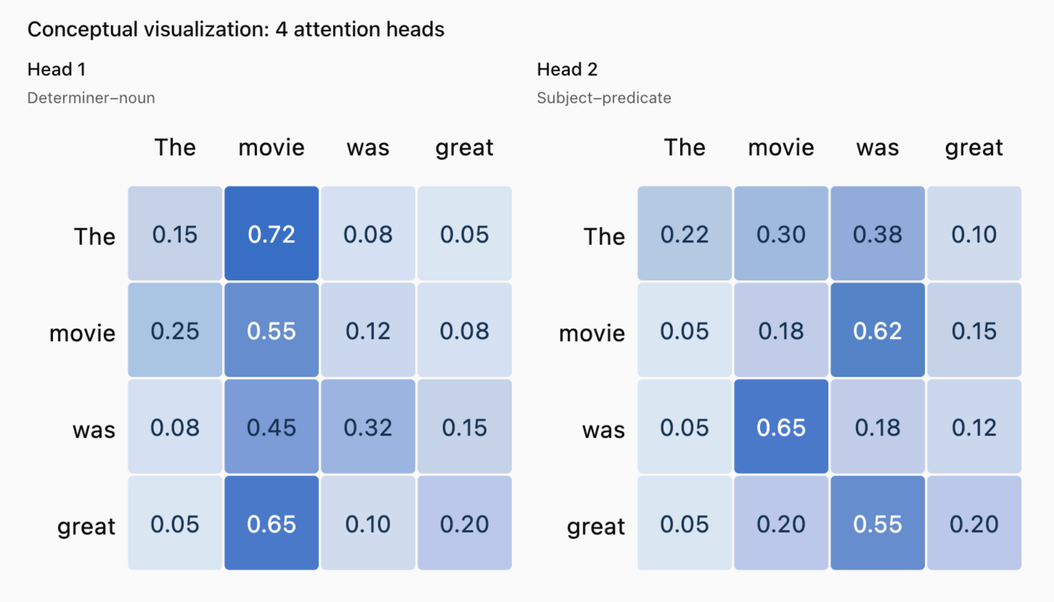
(tokens attending to other tokens)

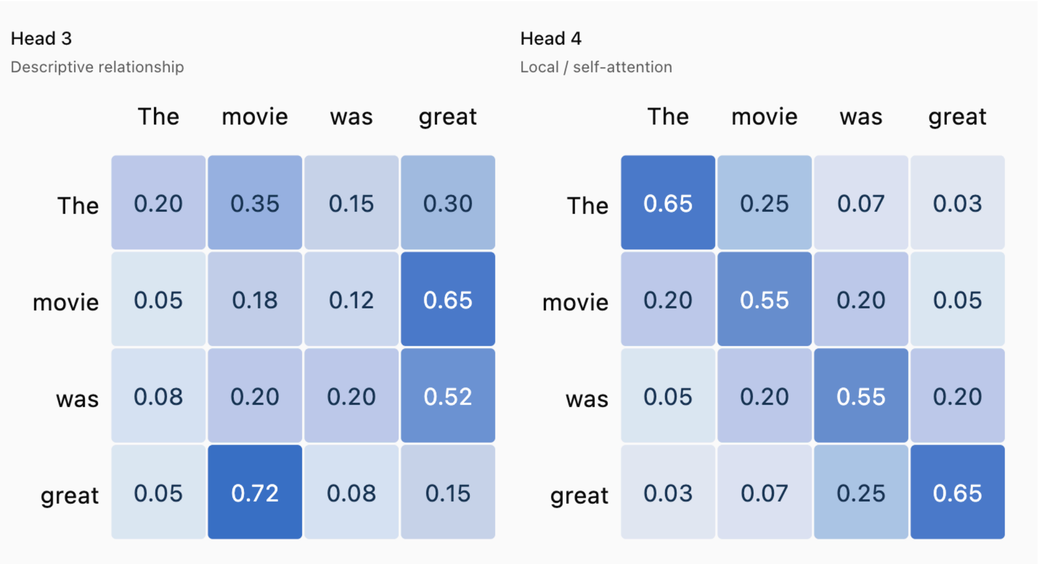
	The	movie	was	great
The	0.25	**0.55**	0.13	0.07
movie	0.10	0.30	0.12	**0.48**
was	0.08	**0.42**	0.35	0.15
great	0.05	**0.60**	0.10	0.25

Queries Keys



Multi-head Self-Attention





- Multiple "heads" use different sets of parameters W_k and V_k to get different attention values + transform vectors

$$\alpha_{h,i,j} = \text{softmax}\left(x_i^\top W_h x_j\right) \qquad x'_{h,i} = \sum_{j=1}^{n} \alpha_{h,i,j} V_h x_j$$

index of attention heads

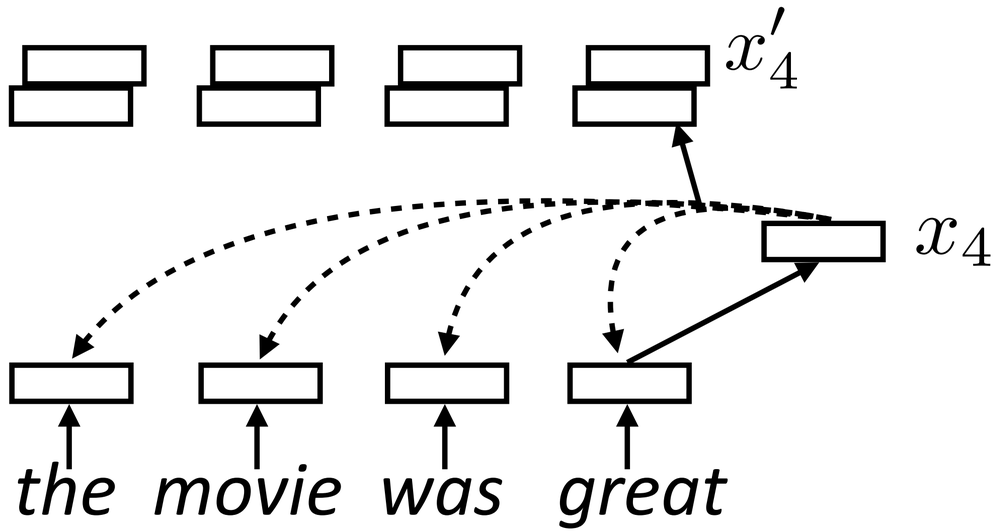
Vaswani et al. (2017)

Multi-head Self-Attention

- Each word forms a "query" which then computes attention over each word

$$\alpha_{i,j} = \text{softmax}(x_i^\top x_j) \quad \text{scalar}$$

$$x'_i = \sum_{j=1}^{n} \alpha_{i,j} x_j \quad \text{vector = sum of scalar * vector}$$



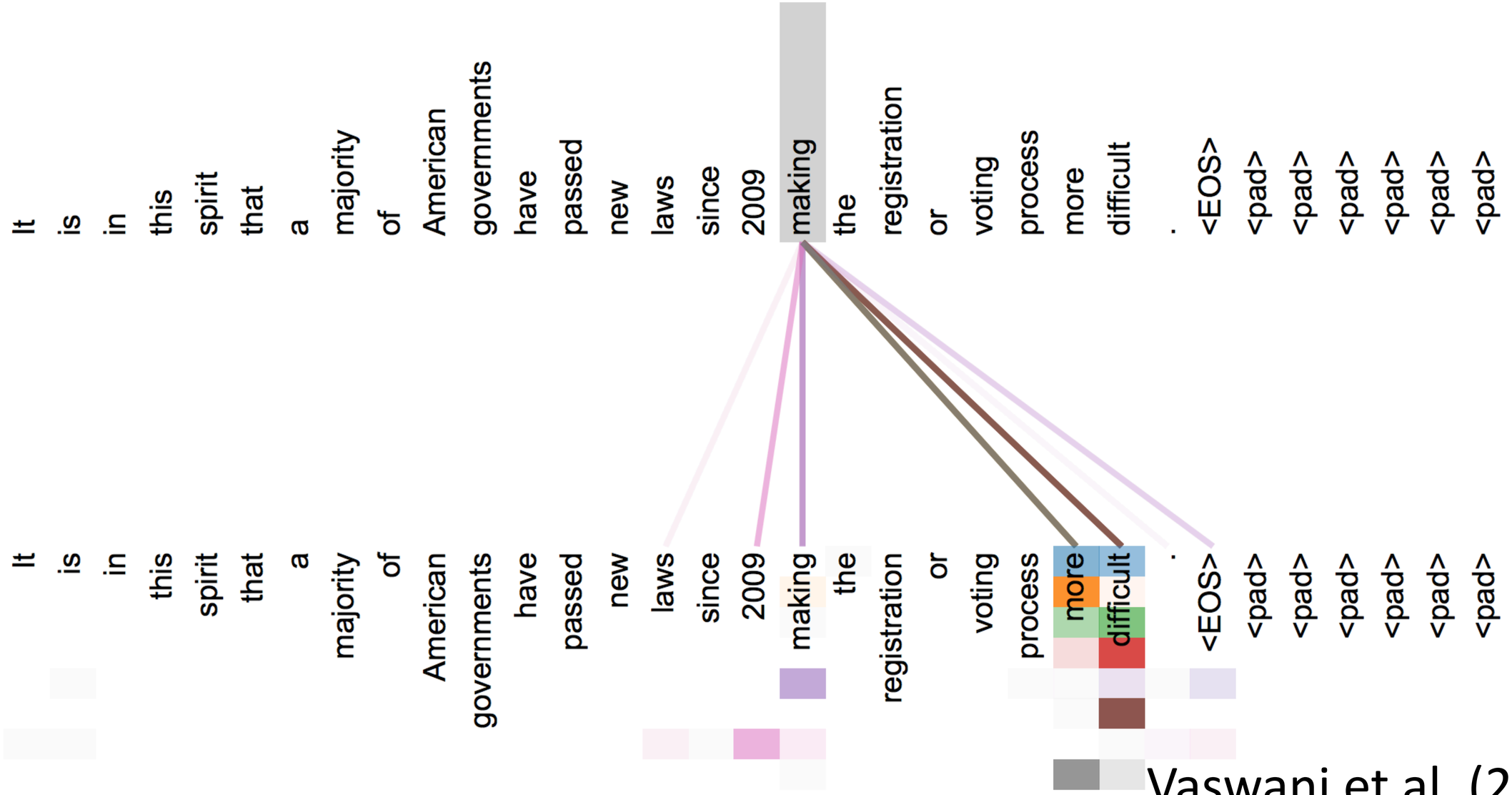
- Multiple "heads" use different sets of parameters W_k and V_k to get different attention values + transform vectors

$$\alpha_{h,i,j} = \text{softmax}\left(x_i^\top W_h x_j\right) \qquad x'_{h,i} = \sum_{j=1}^{n} \alpha_{h,i,j} V_h x_j$$

index of attention heads

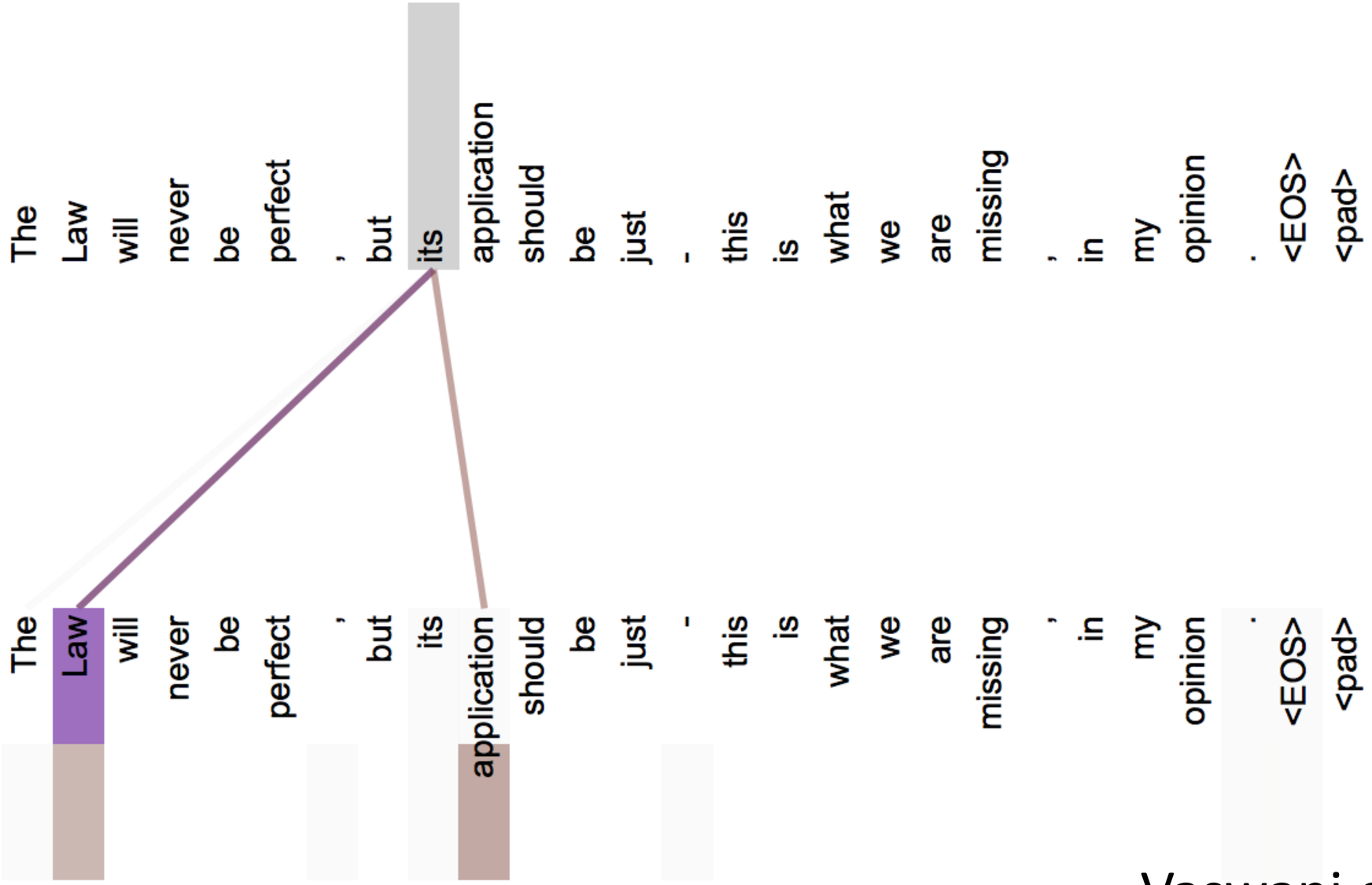
Vaswani et al. (2017)

Visualization



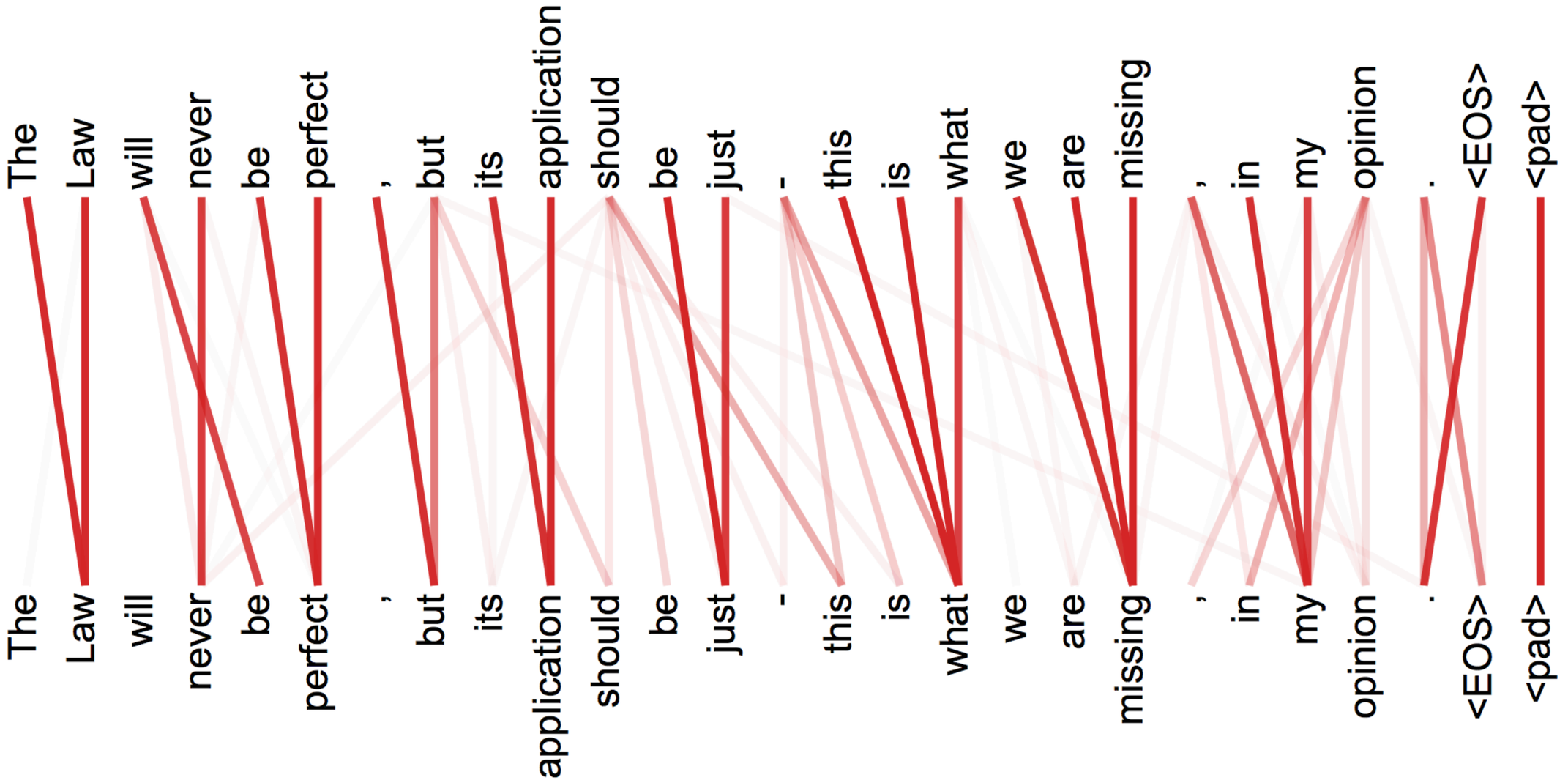
Vaswani et al. (2017)

Visualization



Vaswani et al. (2017)

Visualization



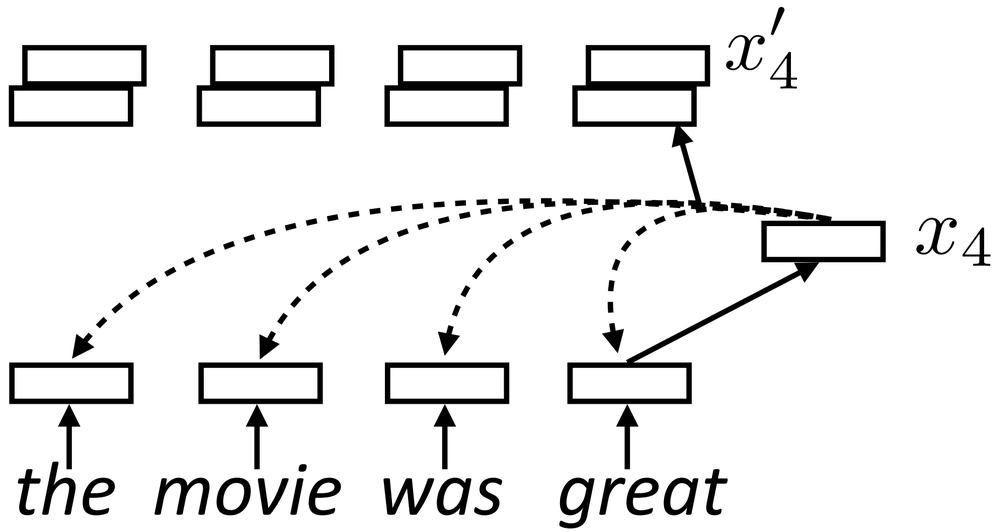
Vaswani et al. (2017)

Multi-head Self-Attention

- Each word forms a "query" which then computes attention over each word

$$\alpha_{i,j} = \text{softmax}(x_i^\top x_j) \quad \text{scalar}$$

$$x'_i = \sum_{j=1}^{n} \alpha_{i,j} x_j \quad \text{vector = sum of scalar * vector}$$



- Multiple "heads" use different sets of parameters W_k and V_k to get different attention values + transform vectors

$$\alpha_{h,i,j} = \text{softmax}\left(x_i^\top W_h x_j\right) \qquad x'_{h,i} = \sum_{j=1}^{n} \alpha_{h,i,j} V_h x_j$$

index of attention heads

Vaswani et al. (2017)

Multi-Head Self Attention

- Multiple “heads” - actual implementation that uses matrix operations
- Let X = [sent len, embedding dim] be the input sentence
- Query $Q = XW^Q$: these are like the **decoder hidden state** in attention
- Keys $K = XW^K$: these control what gets attended to, along with the query
- Values $V = XW^V$: these vectors get summed up to form the output

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

dim of keys

Vaswani et al. (2017)

Multi-Head Self Attention

- Multiple “heads” - actual implementation that uses matrix operations

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

dim of keys

- Each attention head has its own set of learned parameters W^Q, W^K, W^V
- Query vector of token i: $q_i = x_i W^Q$; Key vector of token j: $k_j = x_j W^K$

$$s_{ij} = \frac{q_i k_j^\top}{\sqrt{d_k}} = \frac{(x_i W^Q)(x_j W^K)^\top}{\sqrt{d_k}} = x_i \underbrace{\left(\frac{W^Q (W^K)^\top}{\sqrt{d_k}}\right)}_{W_{\text{bilinear}} \in \mathbb{R}^{d_x \times d_x}} x_j^\top = x_i W_{\text{bilinear}} x_j^\top$$

Vaswani et al. (2017)

Multi-Head Self Attention

- Multiple “heads” - actual implementation that uses matrix operations

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

dim of keys

- Each attention head has its own set of learned parameters W^Q, W^K, W^V
- X as input sentence: $Q = XW^Q, \quad K = X\underbrace{W^K}_{d_x \times d_k}, \quad V = XW^V$

$$\frac{QK^\top}{\sqrt{d_k}} = \frac{(XW^Q)(XW^K)^\top}{\sqrt{d_k}} = X \underbrace{\left(\frac{W^Q(W^K)^\top}{\sqrt{d_k}}\right)}_{W_{\text{bilinear}} \in \mathbb{R}^{d_x \times d_x}} X^\top = XW_{\text{bilinear}}X^\top$$

Vaswani et al. (2017)

Multi-Head Self Attention

- Why scaled by √d_k ?

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

dim of keys & queries

- Query vector of token i: $q_i = x_i W^Q$, Key vector of token j: $k_j = x_j W^K$

- As d_k increases, variance/std. of dot product also grows, causing softmax to become overly peaked

$$s_{ij} = q_i k_j^\top = \sum_{l=1}^{d_k} q_{il} k_{jl}$$

$$\mathbb{E}[s_{ij}] = 0$$
$$\text{Var}(s_{ij}) = d_k$$
$$\text{Std}(s_{ij}) = \sqrt{d_k}$$

Vaswani et al. (2017)

Multi-Head Self Attention

- Why scaled by √d_k ?

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

dim of keys & queries

- Query vector of token i: $q_i = x_i W^Q$, Key vector of token j: $k_j = x_j W^K$

- As d_k increases, variance/std. of dot product also grows, causing softmax to become overly peaked

$$s_{ij} = q_i k_j^\top = \sum_{l=1}^{d_k} q_{il} k_{jl}$$

$$\mathbb{E}[s_{ij}] = 0$$
$$\text{Var}(s_{ij}) = d_k$$
$$\text{Std}(s_{ij}) = \sqrt{d_k}$$

Vaswani et al. (2017)

Multi-Head Self Attention

- Each attention head has its own set of learned parameters W^Q, W^K, W^V
- H attention heads: 1, 2, …, H

$$\text{MultiHead}(X) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H) W^O$$

$$\text{head}_h = \text{softmax}\left(\frac{(XW_h^Q)(XW_h^K)^\top}{\sqrt{d_k}}\right) XW_h^V, \quad h = 1, \ldots, H$$

Vaswani et al. (2017)

Multi-Head Self Attention

- No notion of position of words/tokens in the attention or softmax
- Reordering tokens simply reorders attention weights, leaving each token's weighted sum unchanged.

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

1. Original order

the	movie	was	great
1	2	3	4

2. Attention scores (query = "great")

j	word	score $s_{4j} = q_4 k_j^T / \sqrt{d_k}$
1	the	0.00
2	movie	2.48
3	was	0.69
4	great	1.61

→

3. Softmax (attention weights)

word	α_{4j} (softmax)
the	0.05
movie	0.60
was	0.10
great	0.25

→

4. Output for "great"

$$x'_4 = \sum_{j=1}^{4} \alpha_{4j} x_j$$
$$= 0.05x_{\text{the}} + 0.60x_{\text{movie}} + 0.10x_{\text{was}} + 0.25x_{\text{great}}$$

2. Reordered order

the	great	was	movie
1	2	3	4

2. Attention scores (query = "great")

j	word	score $s_{4j} = q_4 k_j^T / \sqrt{d_k}$
1	the	0.00
2	great	1.61
3	was	0.69
4	movie	2.48

→

3. Softmax (attention weights)

word	α_{4j} (softmax)
the	0.05
great	0.25
was	0.10
movie	0.60

→

4. Output for "great"

$$x'_4 = \sum_{j=1}^{4} \alpha_{4j} x_j$$
$$= 0.05x_{\text{the}} + 0.60x_{\text{movie}} + 0.10x_{\text{was}} + 0.25x_{\text{great}}$$

Vaswani et al. (2017)

Properties of Self-Attention

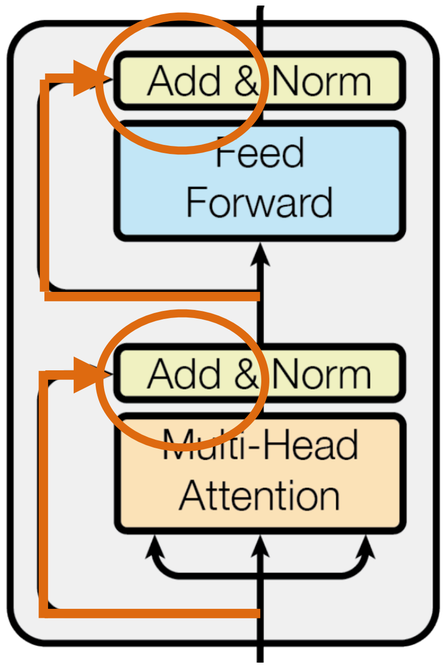
Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

- *n* = sentence length, *d* = hidden dim, *k* = kernel size, *r* = restricted neighborhood size
- **Quadratic complexity**, but O(1) sequential operations (not linear like in RNNs) and O(1) "path" for words to inform each other

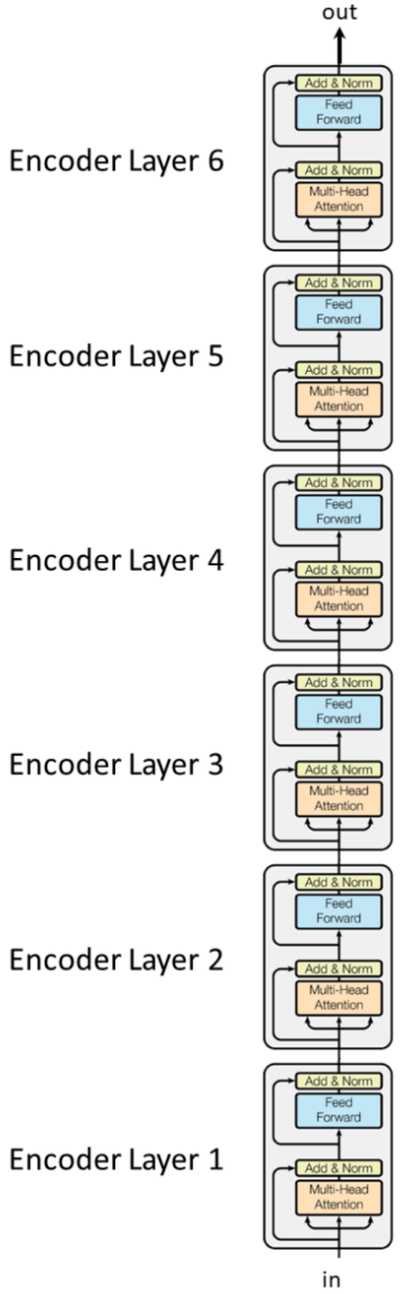
Vaswani et al. (2017)

Transformer

Transformers

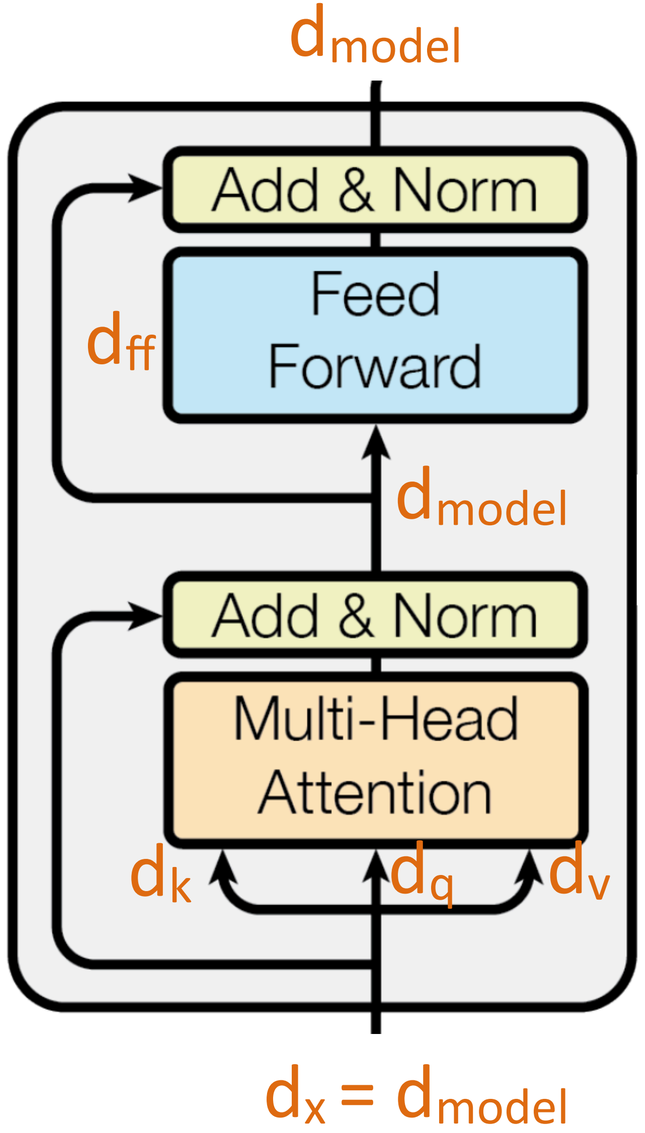


- Alternate multi-head self-attention layers and feedforward layers
- Residual connections let the model "skip" each layer — these are particularly useful for training deep networks



Vaswani et al. (2017)

Transformers



$$X' = \text{LayerNorm}(Z + \text{FFN}(Z))$$

$$Z = \text{LayerNorm}(X + \text{MultiHead}(X))$$

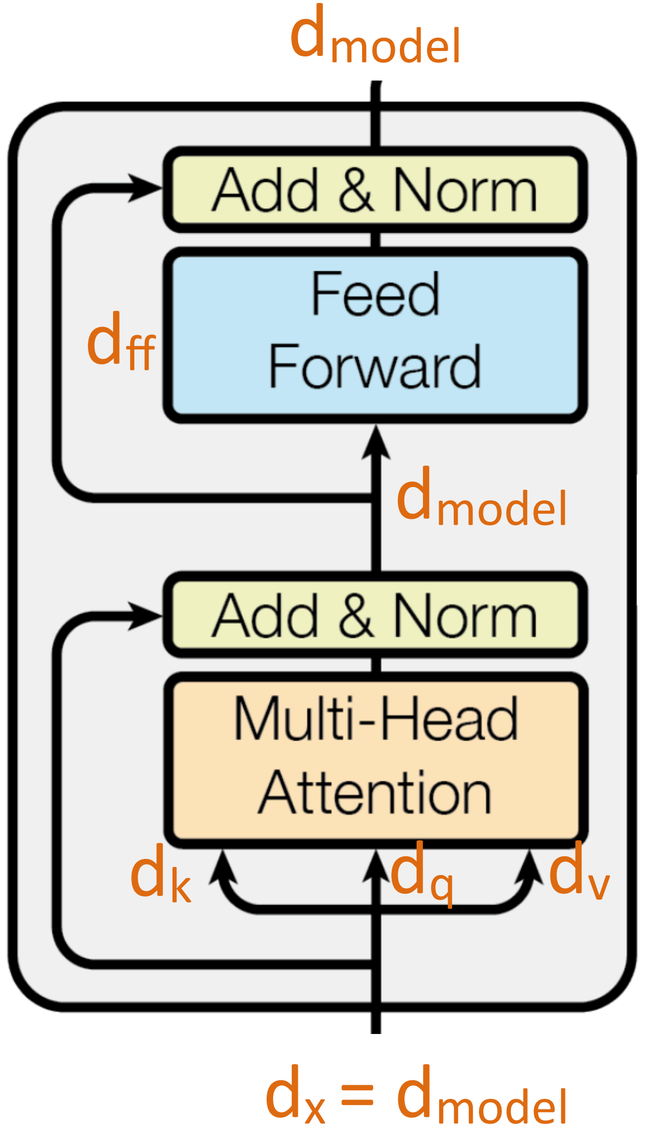
$$X \in \mathbb{R}^{n \times d_{\text{model}}}$$

- Alternate multi-head self-attention layers and feedforward layers that operate over each word individually

← post-LN (original Transformer)

Vaswani et al. (2017)

Transformers



$$X' = \text{LayerNorm}(Z + \text{FFN}(Z))$$

- or $X' = Z + \text{LayerNorm}(\text{FFN}(Z))$

$$Z = \text{LayerNorm}(X + \text{MultiHead}(X))$$

- or $Z = X + \text{LayerNorm}(\text{MultiHead}(X))$

$$X \in \mathbb{R}^{n \times d_{\text{model}}}$$

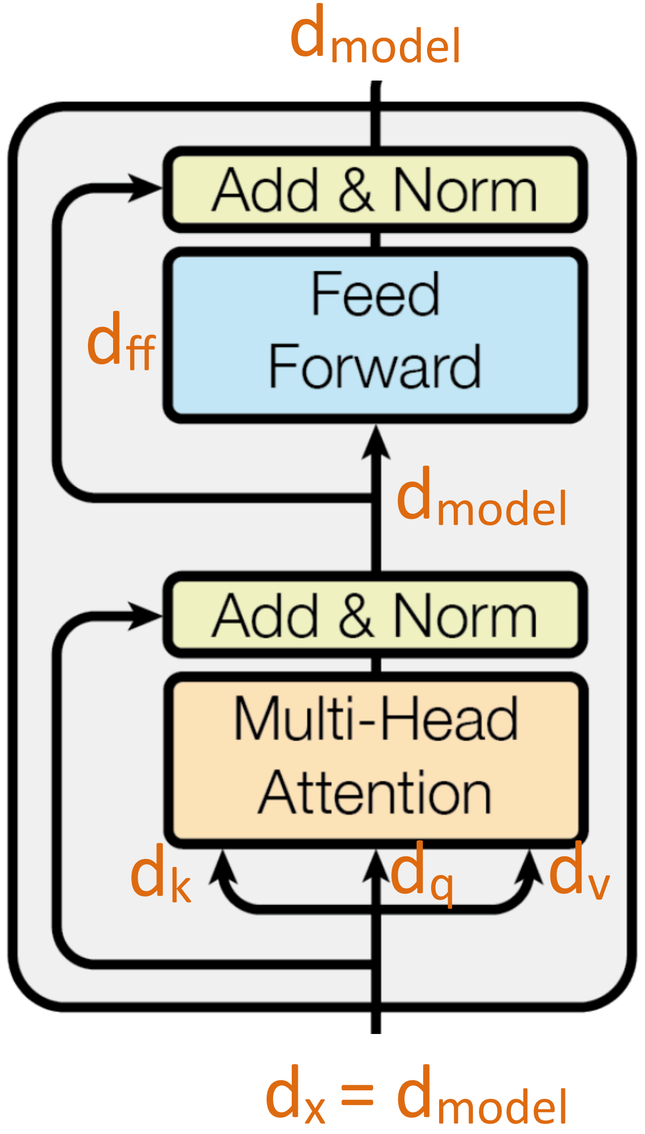
- Alternate multi-head self-attention layers and feedforward layers that operate over each word individually

post-LN (original Transformer)

pre-LN (modern approach)

Vaswani et al. (2017)

Transformers



- Alternate multi-head self-attention layers and feedforward layers (w/ ReLU activation) that operate over each word individually

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

h = 8

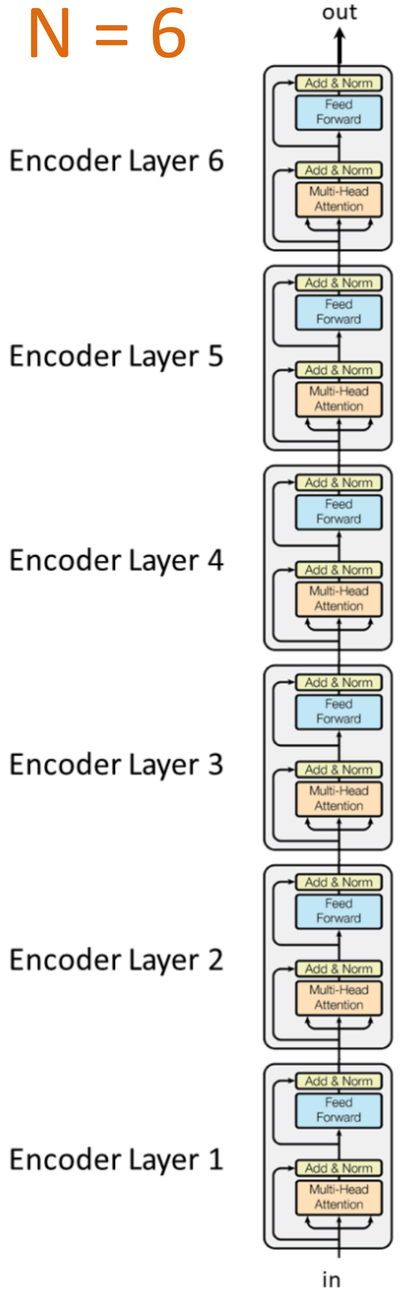
d_x = 512

d_{ff} = 2048

$d_k = d_q = d_{model} / h = 64$

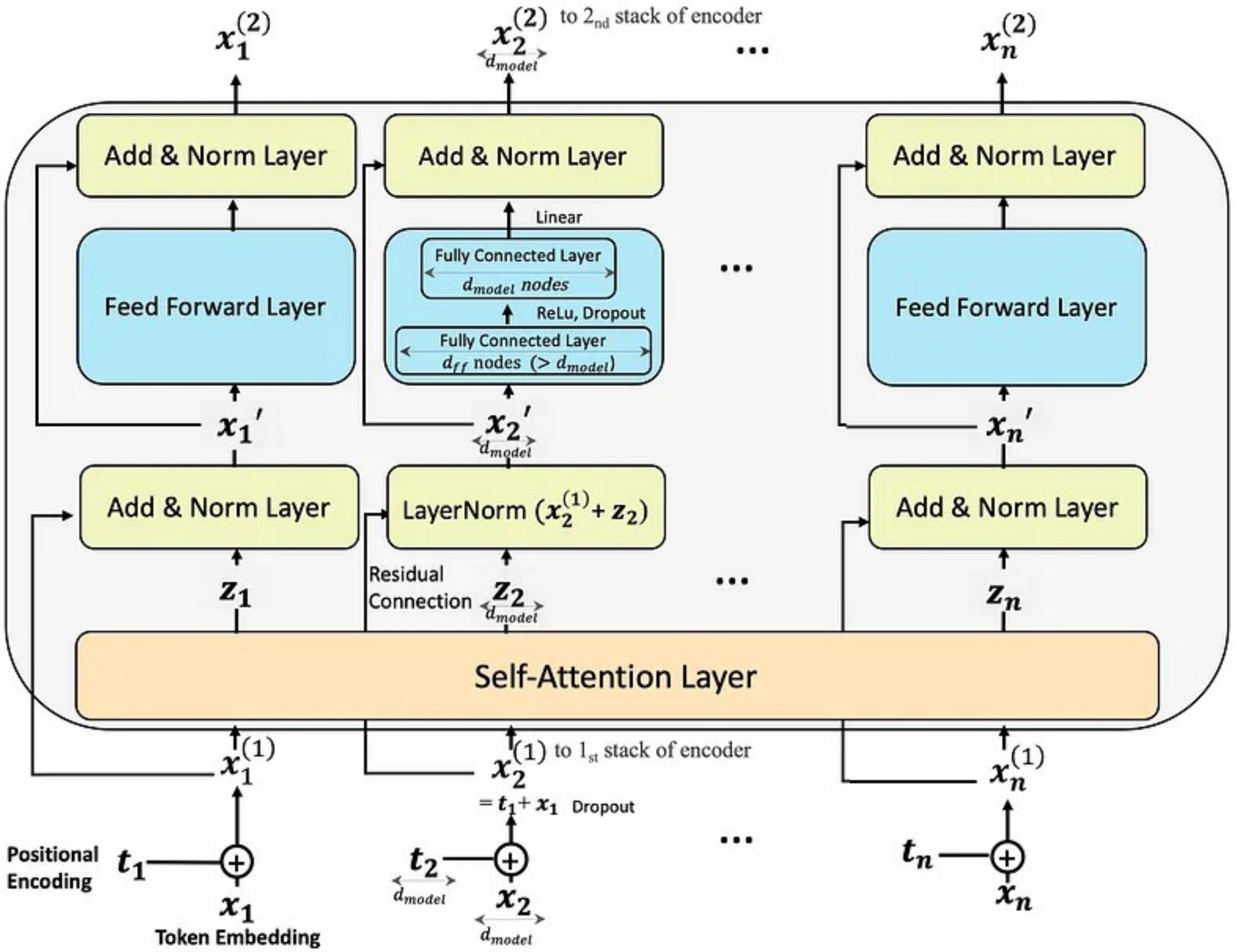
$d_v = d_{model} / h = 64$

N = 6



Vaswani et al. (2017)

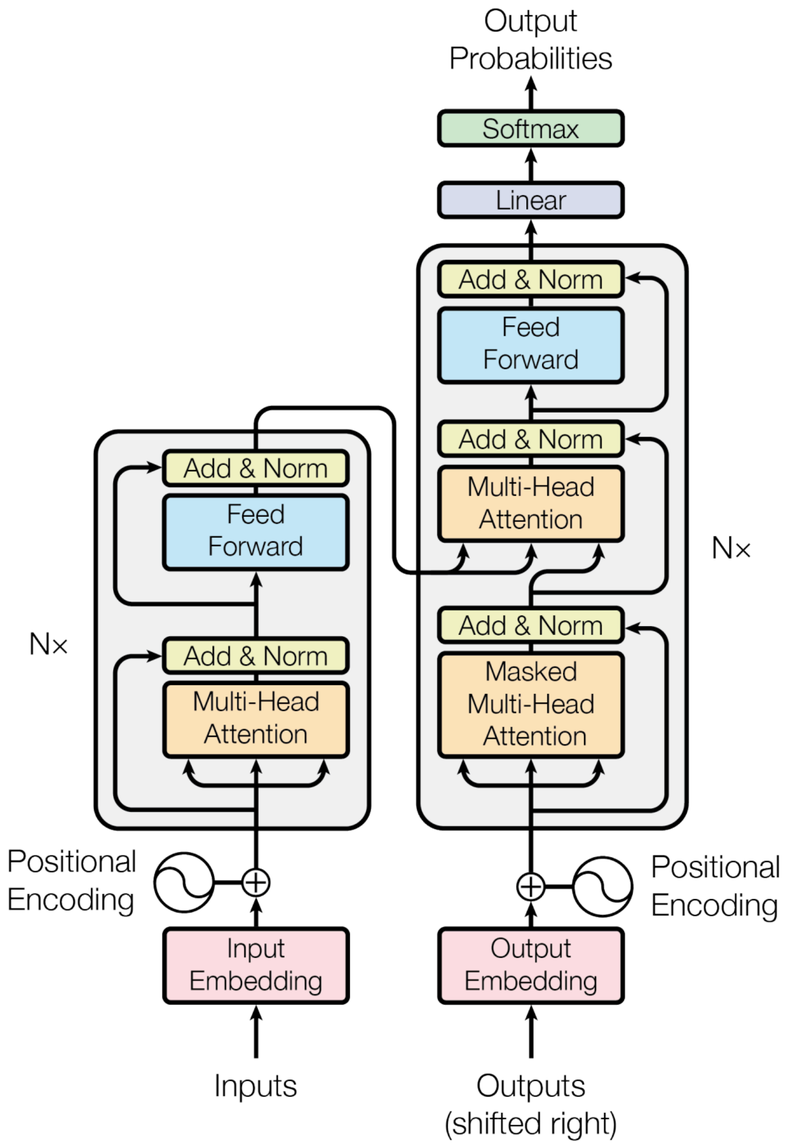
Transformers



- Alternate multi-head self-attention layers and feedforward layers that operate over each word individually

Image Credit: Yule Wang

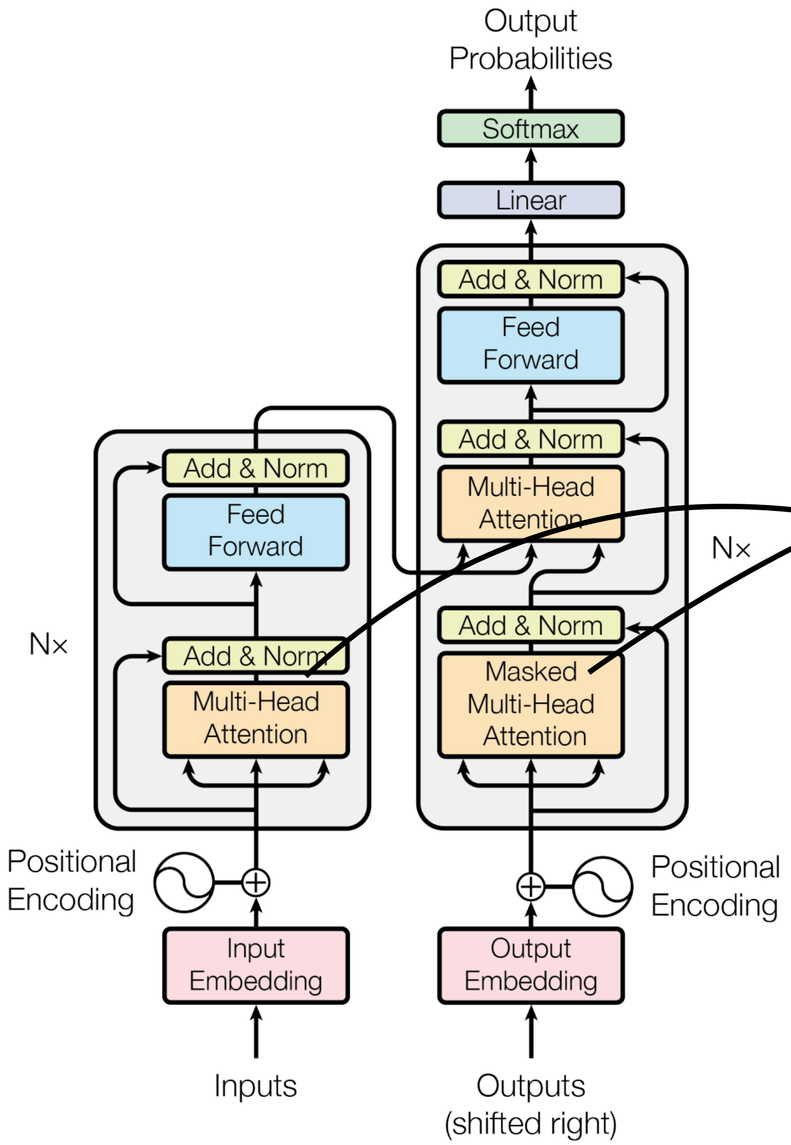
Transformer encoder-decoder

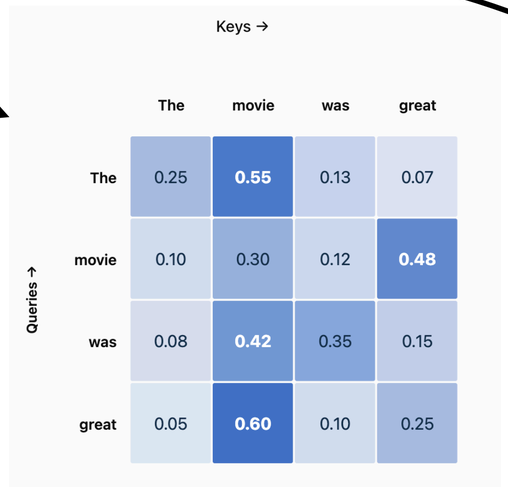


- The original Transformer used an encoder–decoder architecture for machine translation.
- Transformers can be used as encoders, decoders, or both, similar to LSTMs and CNNs.

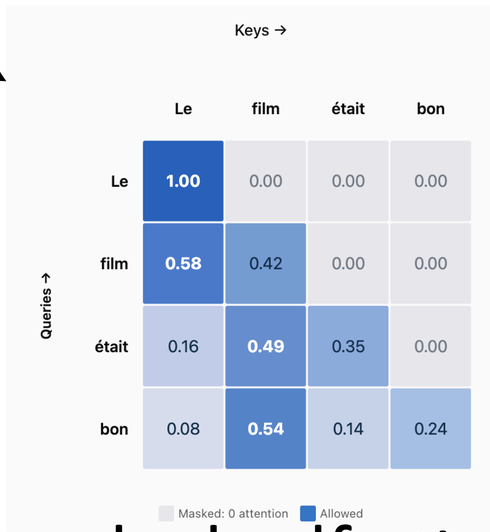
Transformer encoder-decoder

- The original Transformer used an encoder–decoder architecture for machine translation.
- Transformers can be used as encoders, decoders, or both, similar to LSTMs and CNNs.





self-attention (encoder)



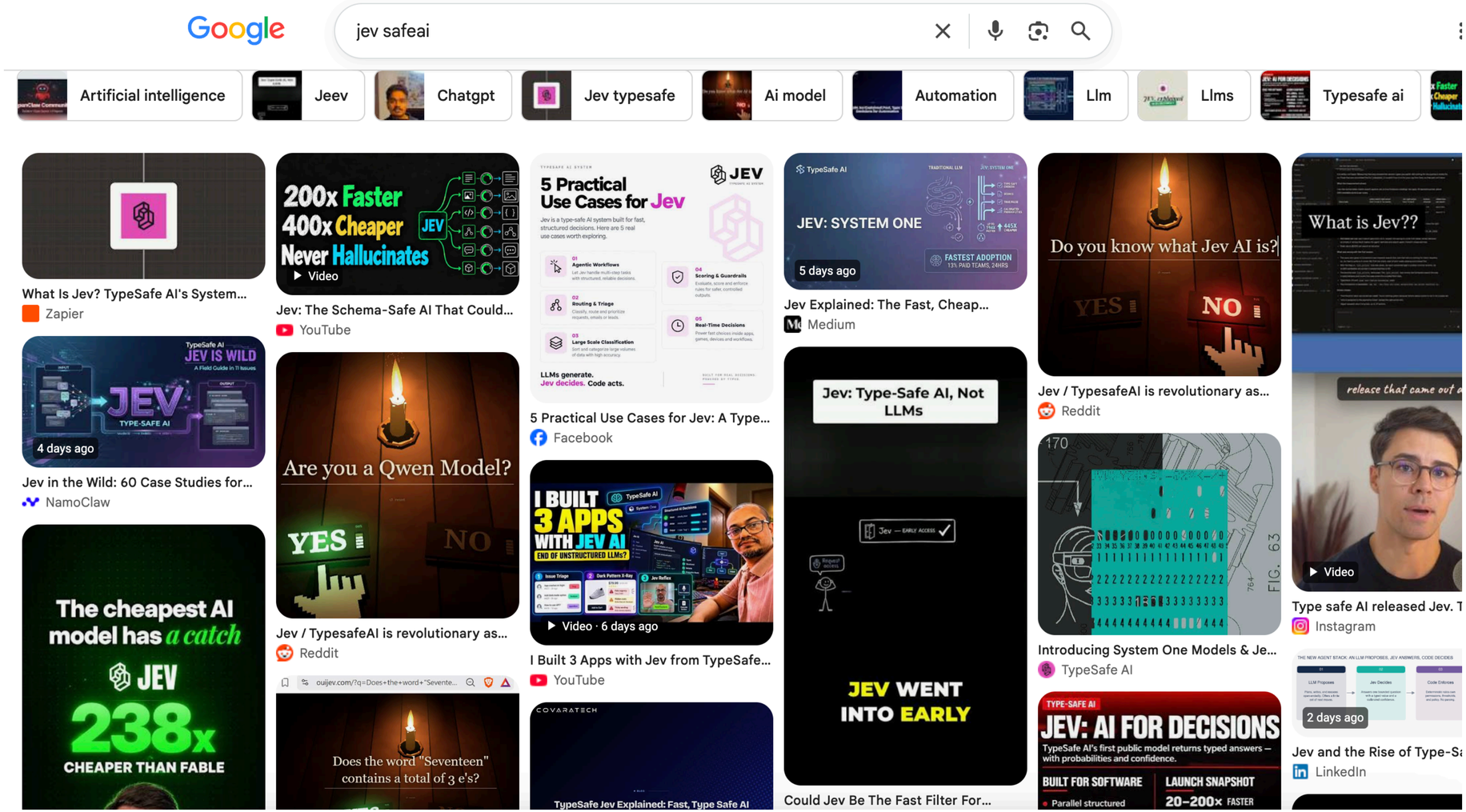
masked self-attention (decoder)

Masked Language Models

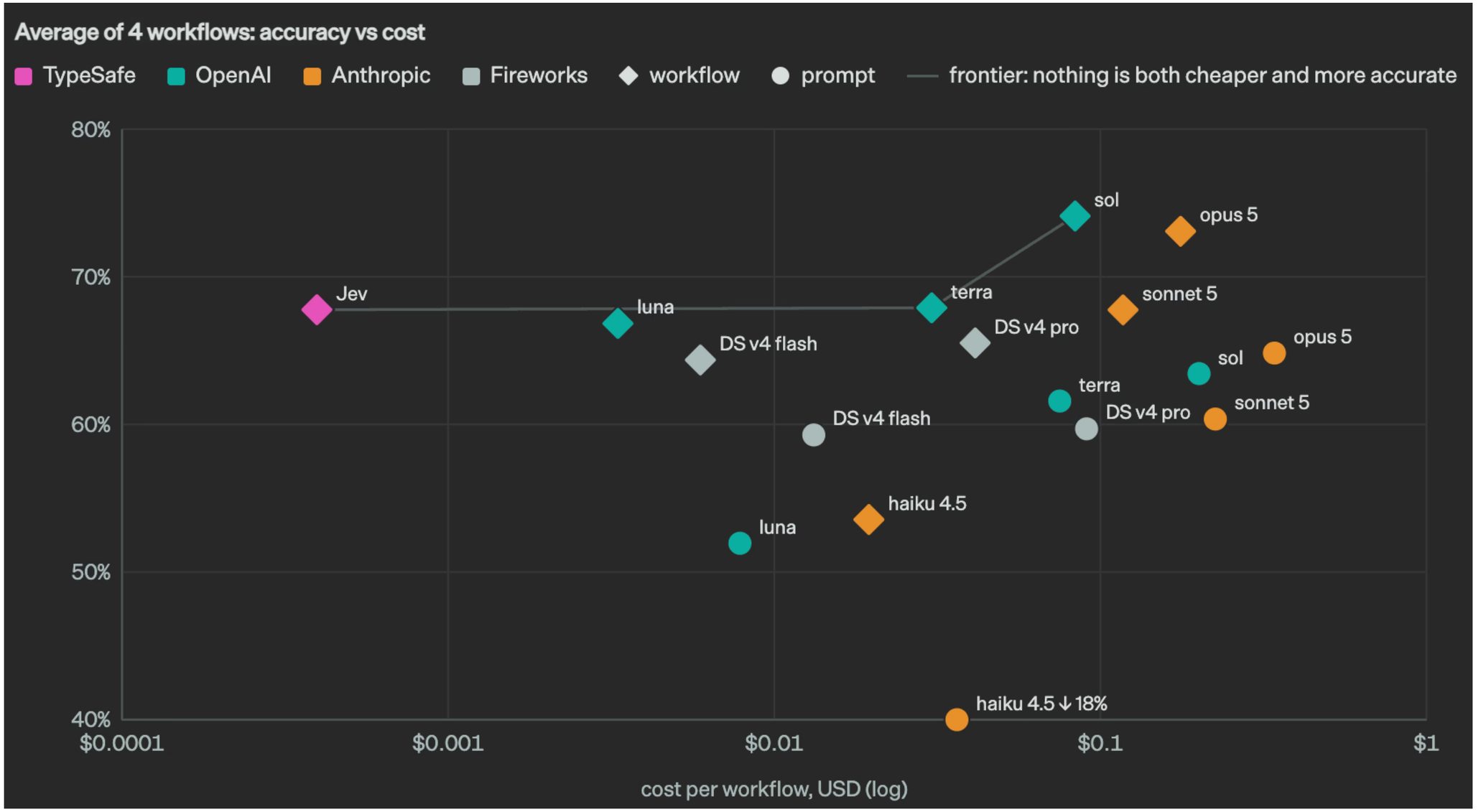
Wei Xu

(many slides from Greg Durrett)

What’s new? Jev model - released 9/15/2026



What’s new? Jev model - released 9/15/2026

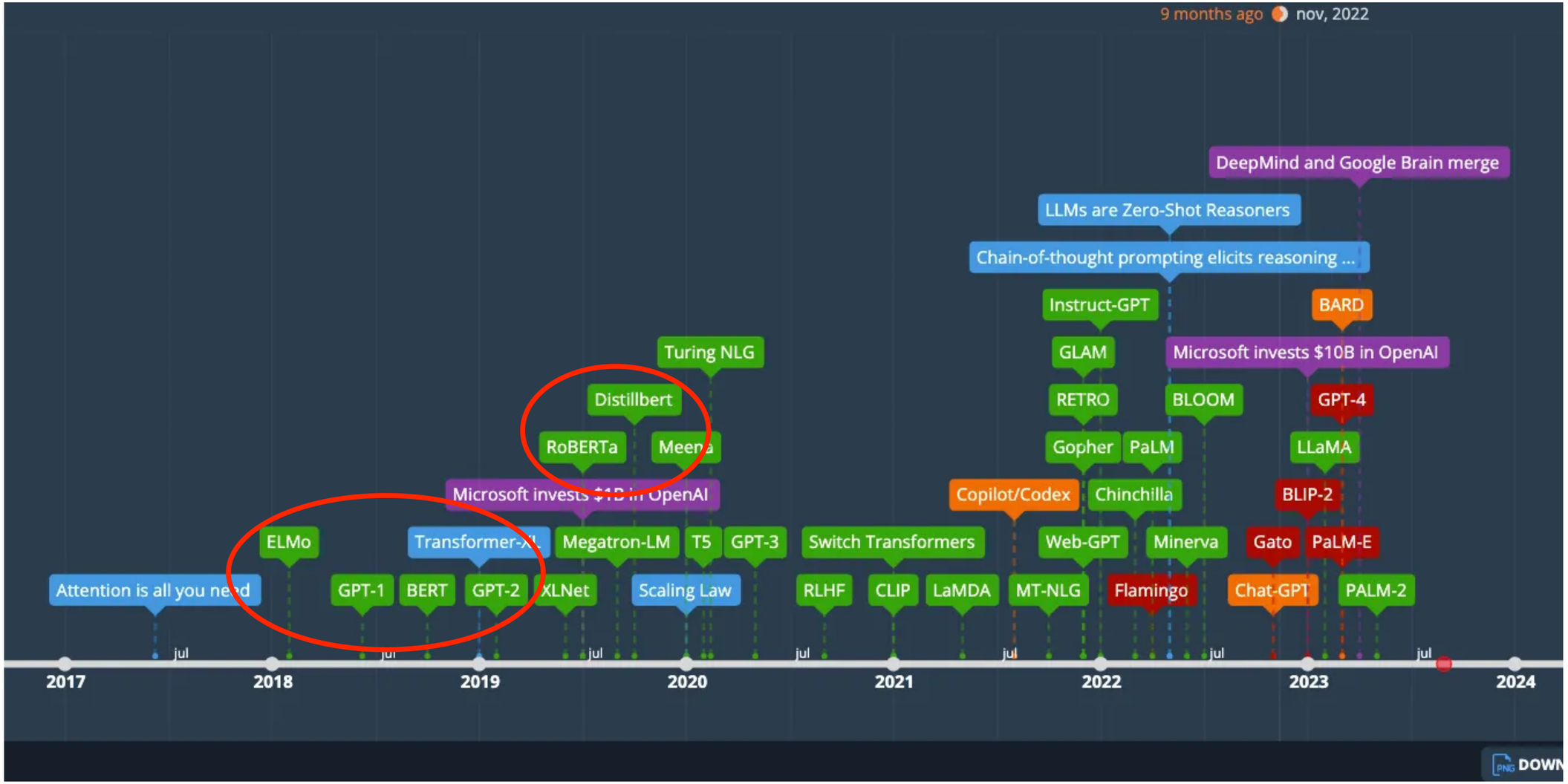


This Lecture

- Encoder-based LMs (e.g., ELMo, BERT)

- Next Lectures:
  - Decoder-based LMs (e.g., GPT-1/GPT-2/GPT-3)
  - Encoder-decoder LMs (e.g., BART, T5)
  - Knowledge Distillation (e.g., DistillBERT)

The beginning of LLMs

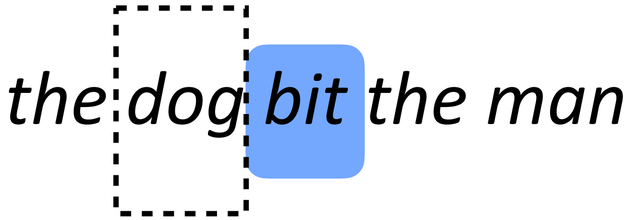


Readings

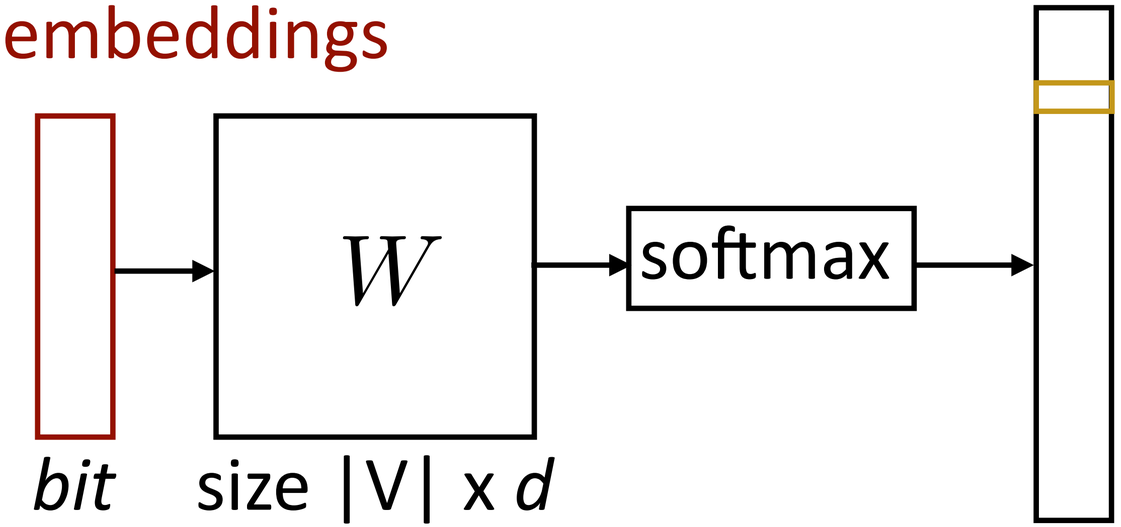
- Readings —
  - J+M 9
  - ELMo by Peters et al.
    https://aclanthology.org/N18-1202.pdf
  - BERT by Devlin et al.
    https://aclanthology.org/N19-1423.pdf

Recall: word2vec (Skip-Gram)

- Predict one word of context from word



d-dimensional
word embeddings



gold label = *dog*

$$P(w'|w) = \text{softmax}(We(w))$$

- Another training example: *bit* -> *the*
- Parameters: *d* x |V| vectors, |V| x *d* output parameters (W) (also usable as vectors!)

Mikolov et al. (2013)

ELMo

(released 2018/2, early version 2017/10)

ELMo

Deep contextualized word representations

Matthew E. Peters[†], **Mark Neumann**[†], **Mohit Iyyer**[†], **Matt Gardner**[†],
{matthewp,markn,mohiti,mattg}@allenai.org

Christopher Clark[*], **Kenton Lee**[*], **Luke Zettlemoyer**[†*]
{csquared,kentonl,lsz}@cs.washington.edu

[†]Allen Institute for Artificial Intelligence
[*]Paul G. Allen School of Computer Science & Engineering, University of Washington

Abstract

We introduce a new type of *deep contextualized* word representation that models both (1) complex characteristics of word use (e.g., syntax and semantics), and (2) how these uses vary across linguistic contexts (i.e., to model polysemy). Our word vectors are learned functions of the internal states of a deep bidirectional language model (biLM), which is pre-trained on a large text corpus. We show that these representations can be easily added to existing models and significantly improve the state of the art across six challenging NLP problems, including question answering, textual entailment and sentiment analysis. We also present an analysis showing that exposing the deep internals of the pre-trained network is crucial, allowing downstream models to mix different types of semi-supervision signals.

1 Introduction

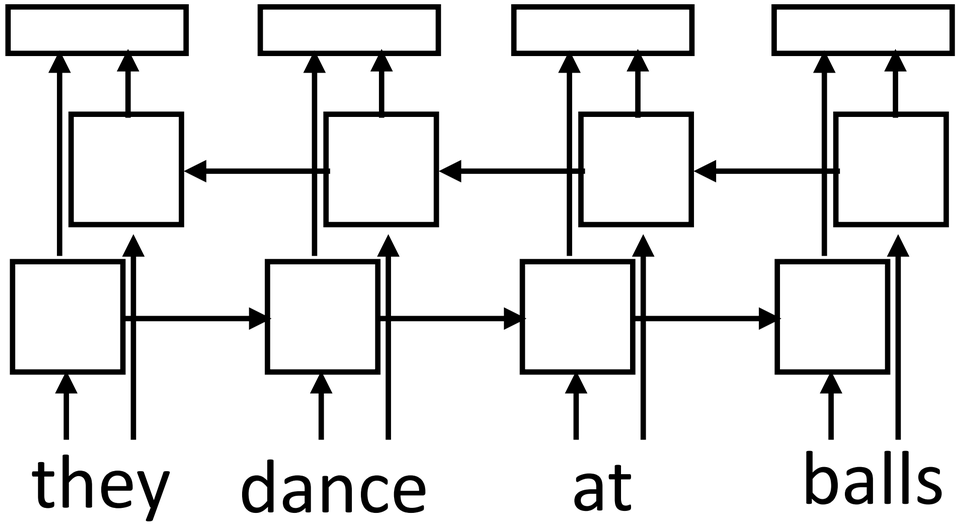
Pre-trained word representations (Mikolov et al., 2013; Pennington et al., 2014) are a key component in many neural language understanding models. However, learning high quality representations can be challenging. They should ideally model both (1) complex characteristics of word use (e.g., syntax and semantics), and (2) how these uses vary across linguistic contexts (i.e., to model polysemy). In this paper, we introduce a general purpose pre-trained lan-

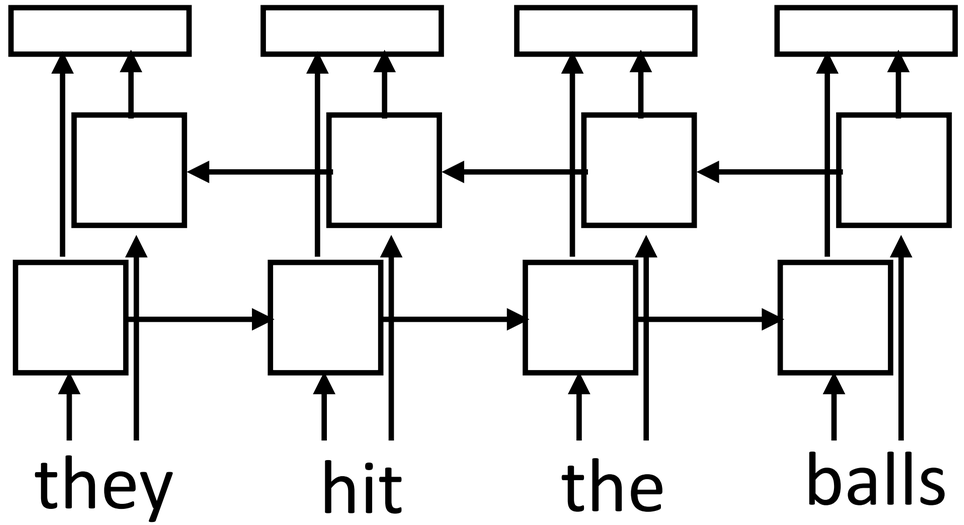
guage model (LM) objective on a large text corpus. For this reason, we call them ELMo (Embeddings from Language Models) representations. Unlike previous approaches for learning contextualized word vectors (Peters et al., 2017; McCann et al., 2017), ELMo representations are deep, in the sense that they are a function of all of the internal layers of the biLM. More specifically, we learn a linear combination of the vectors stacked above each input word for each end task, which markedly improves performance over just using the top LSTM layer.

Combining the internal states in this manner allows for very rich word representations. Using intrinsic evaluations, we show that the higher-level LSTM states capture context-dependent aspects of word meaning (e.g., they can be used without modification to perform well on supervised word sense disambiguation tasks) while lower-level states model aspects of syntax (e.g., they can be used to do part-of-speech tagging). Simultaneously exposing all of these signals is highly bene-

Context-dependent Embeddings

- How to handle different word senses? One vector for *balls*



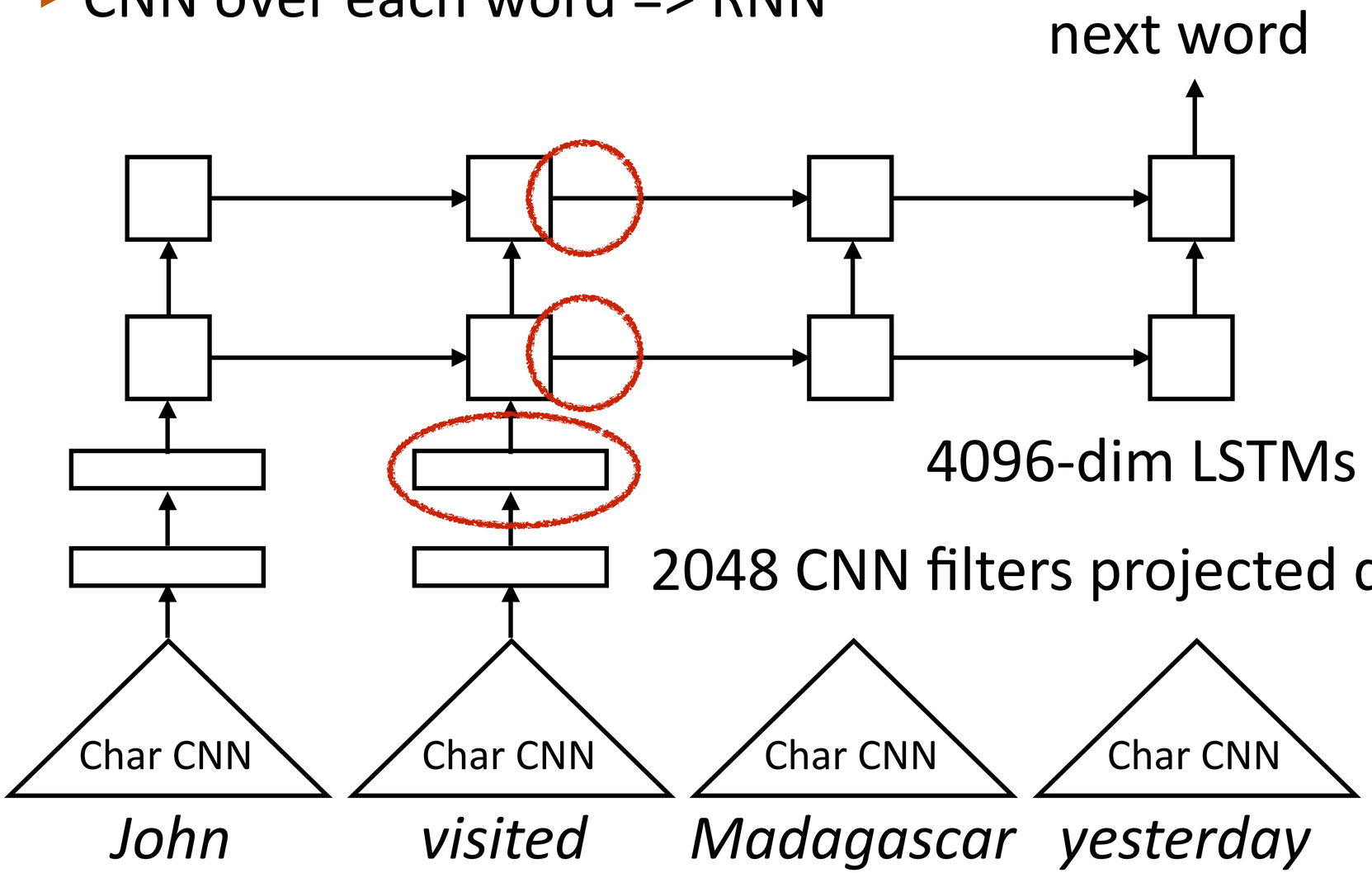


- Train a neural language model to predict the next word given previous words in the sentence, use its internal representations as word vectors
- *Context-sensitive* word embeddings: depend on rest of the sentence
- *Huge* improvements across nearly all NLP tasks over word2vec & GloVe

ELMo - Peters et al. (2018)

ELMo

- CNN over each word => RNN



Representation of *visited* (plus vectors from backwards LM)

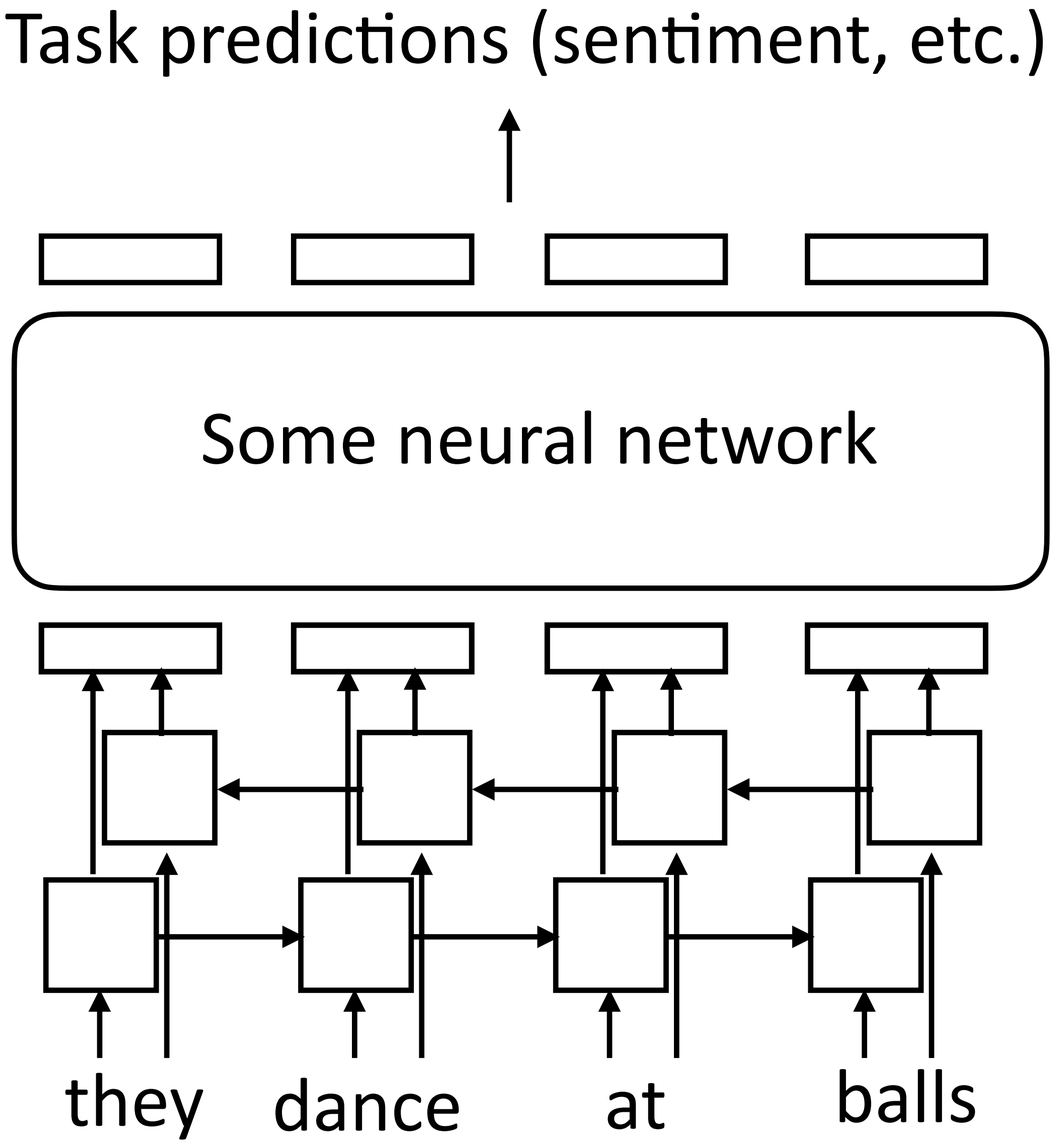
4096-dim LSTMs w/ 512-dim projections

2048 CNN filters projected down to 512-dim

Peters et al. (2018)

How to apply ELMo?

- Take those embeddings and feed them into whatever architecture you want to use for your task
- *Frozen* embeddings: update the weights of your network but keep ELMo's parameters frozen
- *Fine-tuning*: backpropagate all the way into ELMo when training your model



Results: Frozen ELMo

Task	Previous SOTA		Our baseline	ELMo + baseline	Increase (absolute/ relative)
SQuAD	Liu et al. (2017)	84.4	81.1	85.8	4.7 / 24.9%
SNLI	Chen et al. (2017)	88.6	88.0	88.7 ± 0.17	0.7 / 5.8%
SRL	He et al. (2017)	81.7	81.4	84.6	3.2 / 17.2%
Coref	Lee et al. (2017)	67.2	67.2	70.4	3.2 / 9.8%
NER	Peters et al. (2017)	91.93 ± 0.19	90.15	92.22 ± 0.10	2.06 / 21%
SST-5	McCann et al. (2017)	53.7	51.4	54.7 ± 0.5	3.3 / 6.8%

- Massive improvements across 5 benchmark datasets: question answering, natural language inference, semantic role labeling, coreference resolution, named entity recognition, and sentiment analysis

Peters et al. (2018)

ELMo

- Key idea: language models can allow us to form useful word representations in the same way word2vec did
- Take a powerful language model, train it on large amounts of data, then use those contextualized representations in downstream tasks
  - Data: Wikipedia, books, crawled stuff from the web, …

Peters et al. (2018)

OpenReview

OpenReview.net Search OpenReview... Login

Deep contextualized word representations

Matthew E Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, Luke Zettlemoyer

02 Feb 2018 (modified: 02 Mar 2025) ICLR 2018 Conference Blind Submission Readers: Everyone Show Bibtex Show Revisions

Keywords: representation learning, contextualized word embeddings

TL;DR: We introduce a new type of deep contextualized word representation that significantly improves the state of the art for a range of challenging NLP tasks.

Abstract: We introduce a new type of deep contextualized word representation that models both (1) complex characteristics of word use (e.g., syntax and semantics), and (2) how these uses vary across linguistic contexts (i.e., to model polysemy). Our word vectors are learned functions of the internal states of a deep bidirectional language model (biLM), which is pretrained on a large text corpus. We show that these representations can be easily added to existing models and significantly improve the state of the art across six challenging NLP problems, including question answering, textual entailment and sentiment analysis. We also present an analysis showing that exposing the deep internals of the pretrained network is crucial, allowing downstream models to mix different types of semi-supervision signals.

Community Implementations: 43 code implementations

Withdrawal: Confirmed

Reply Type: all Author: everybody Visible To: all readers Hidden From: nobody **17 Replies**

[−] NLP conference

ICLR 2018 Conference Paper759 Authors

02 Feb 2018, 21:17 ICLR 2018 Conference Paper759 Official Comment Readers: Everyone Show Revisions

Comment: As suggested by the meta reviewer, the empirical results in this paper are better suited for a NLP conference then ICLR. As a result, we are withdrawing it from ICLR. Thank you to the Chairs for your consideration and paper acceptance.

[−] ICLR 2018 Conference Acceptance Decision

ICLR 2018 Conference Program Chairs

29 Jan 2018, 13:11 (modified: 29 Jan 2018, 16:08) ICLR 2018 Conference Acceptance Decision Readers: Everyone Show Revisions

Decision: Accept (Poster)

Comment: This is a good paper that presents state-of-the-art results on a number of challenging NLP tasks. The idea is fairly simple and clean, I therefore expect it to get adopted in the community. It also seems to work across several tasks, which is nice. At the same time it is fairly simple (train an LSTM language model and use representations from all levels of the LSTM in the input or output layer of a supervised task of interest) and hence may be more appropriate for an NLP conference than ICLR?

It is a good paper and it will get cited even though the ML contributions are modest. Accept due to the strong empirical results.

BERT

BERT

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin **Ming-Wei Chang** **Kenton Lee** **Kristina Toutanova**
Google AI Language
{jacobdevlin,mingweichang,kentonl,kristout}@google.com

arXiv:1810.04805v2 [cs.CL] 24 May 2019

Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

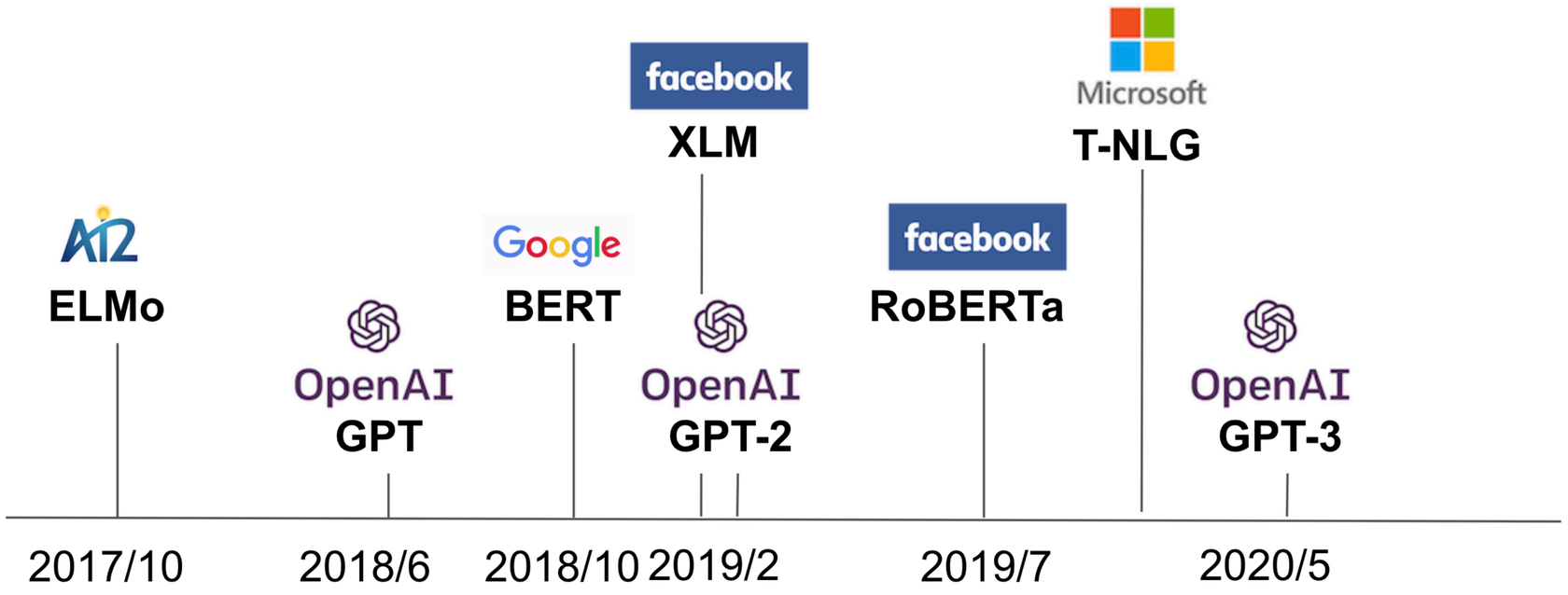
There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers

Contextual Word Embeddings

- AI2 released ELMo in spring 2018, GPT (transformer-based) was released in summer 2018, BERT (transformer-based) came out October 2018

- aka Pre-trained Language Models



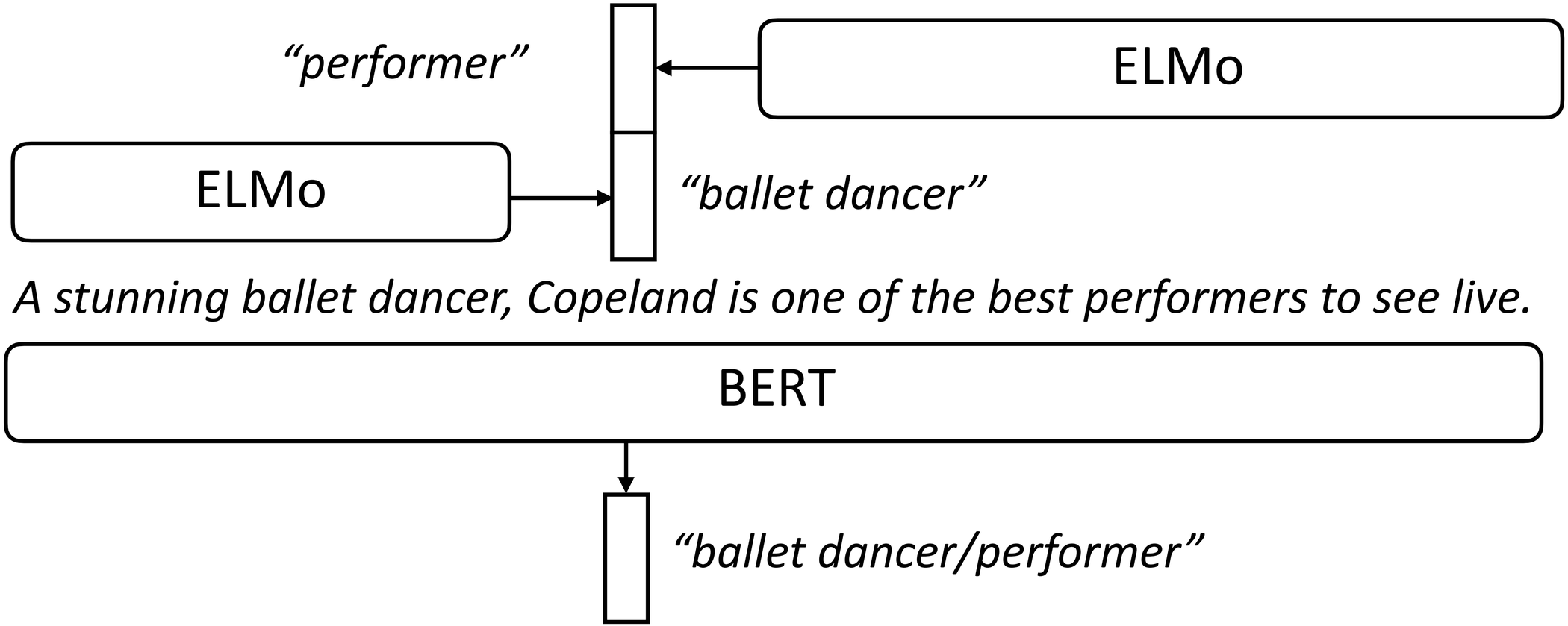
and many more …

Contextual Word Embeddings

- AI2 released ELMo in spring 2018, GPT (transformer-based) was released in summer 2018, BERT (transformer-based) came out October 2018
- aka Pre-trained Language Models
- BERT's four major changes compared to ELMo:
  - Transformers instead of LSTMs (transformers in GPT as well)
  - "Truly" Bidirectional <=> Masked LM objective instead of standard LM
  - Fine-tune instead of freeze at test time
  - Uses word pieces (subword tokenization)

BERT

- ELMo is a unidirectional model: we can concatenate two unidirectional models, but is this the right thing to do?



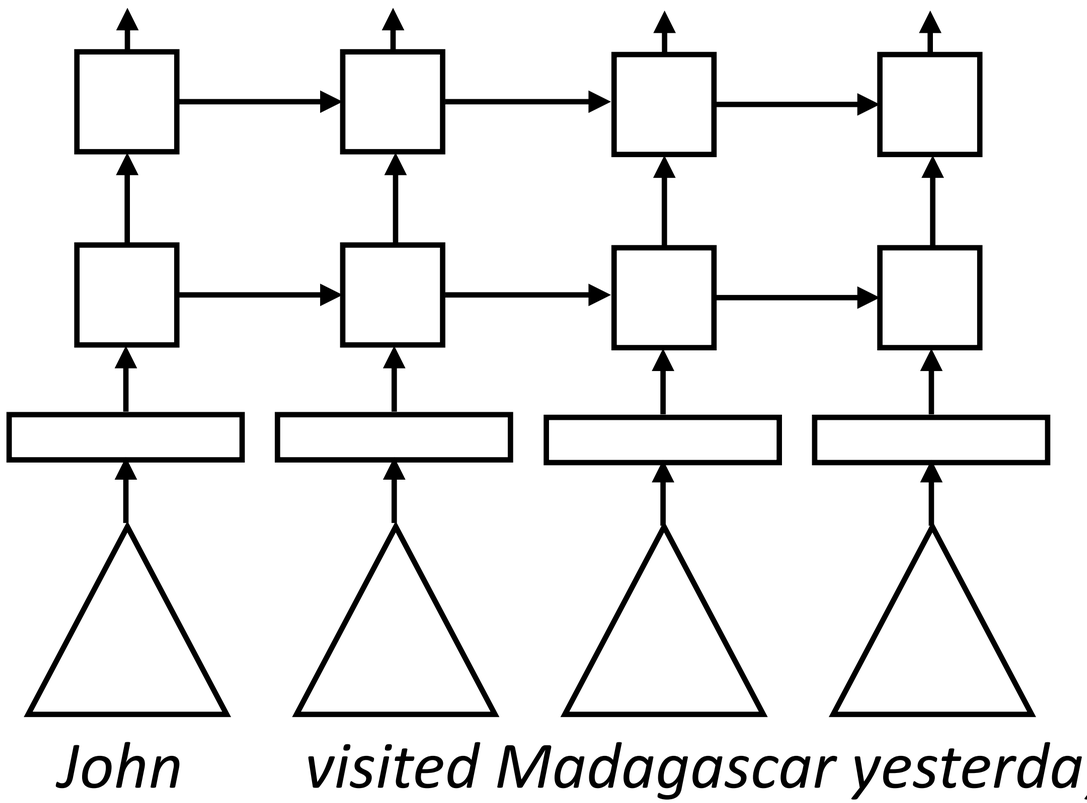
- ELMo looks at each direction in isolation; BERT looks at them jointly

Devlin et al. (2019)

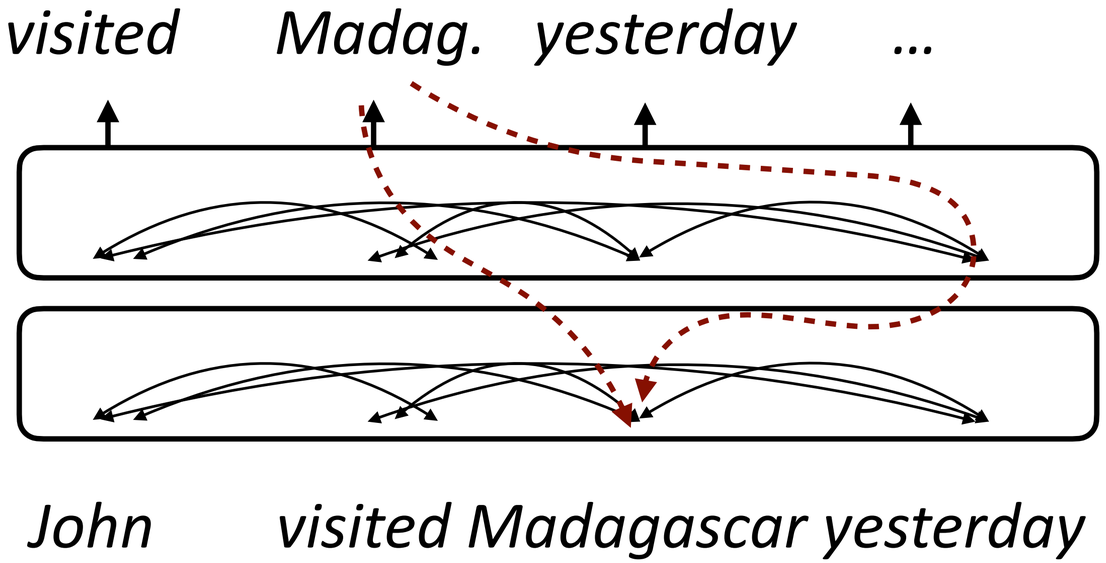
BERT

- How to learn a “deeply bidirectional” model? What happens if we just replace an LSTM with a transformer?

ELMo (Language Modeling)



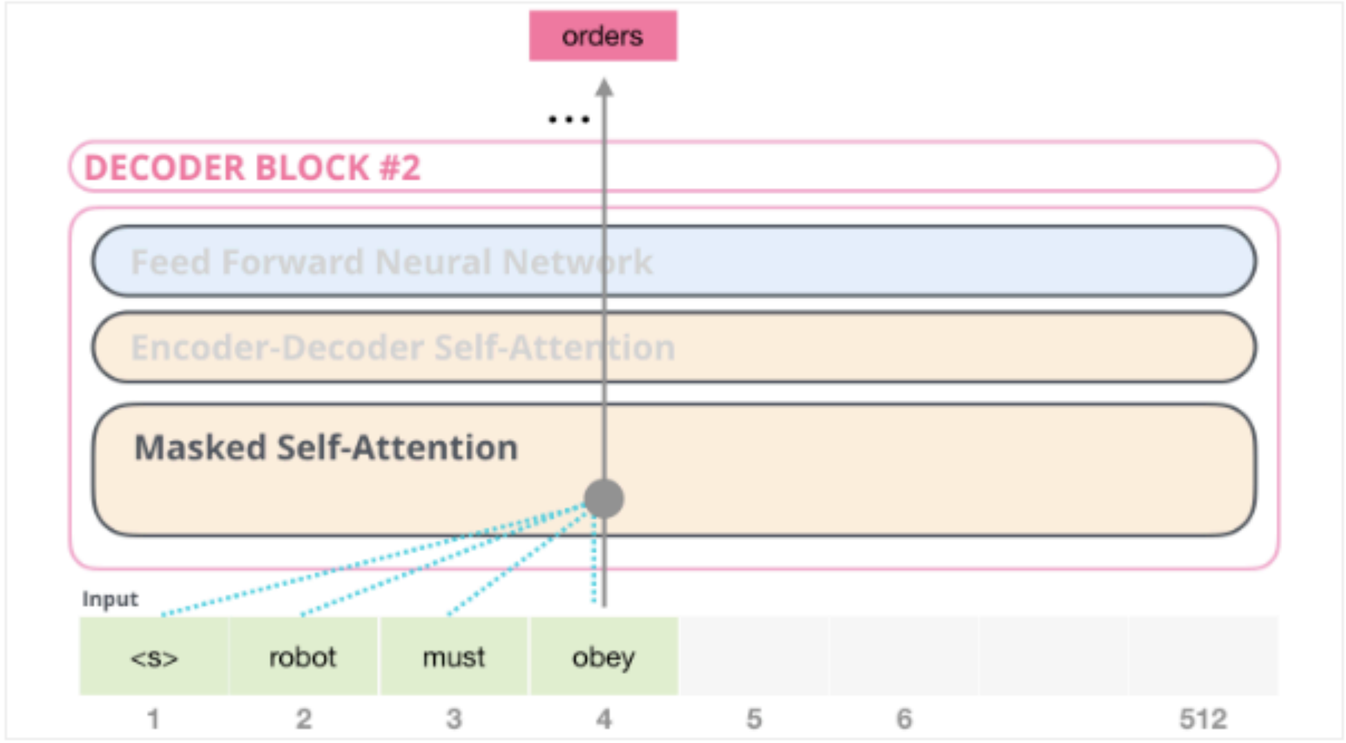
BERT



- Can do this by “one-sided” Transformer (masked self-attention), but “two-sided” Transformer encoder can cheat

(Preview) GPT

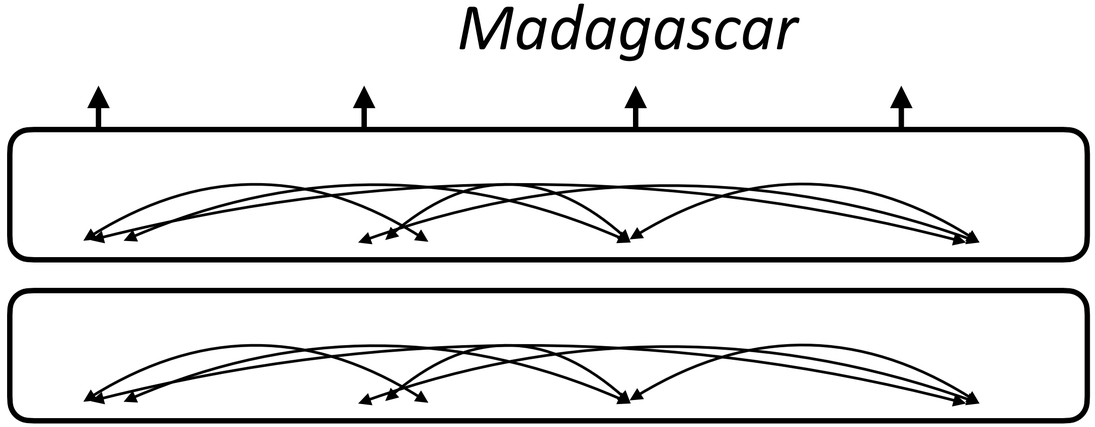
- Transformer with masked self-attention: each token can only attend to past tokens



Masked Language Modeling (MLM)

- How to prevent cheating? Next word prediction fundamentally doesn't work for bidirectional models, instead do *masked language modeling*

- BERT formula: take a chunk of text, predict 15% of the tokens
  - For 80% (of the 15%), replace the input token with [MASK]
  - For 10%, replace w/random
  - For 10%, keep same



John *visited* [MASK] *yesterday*

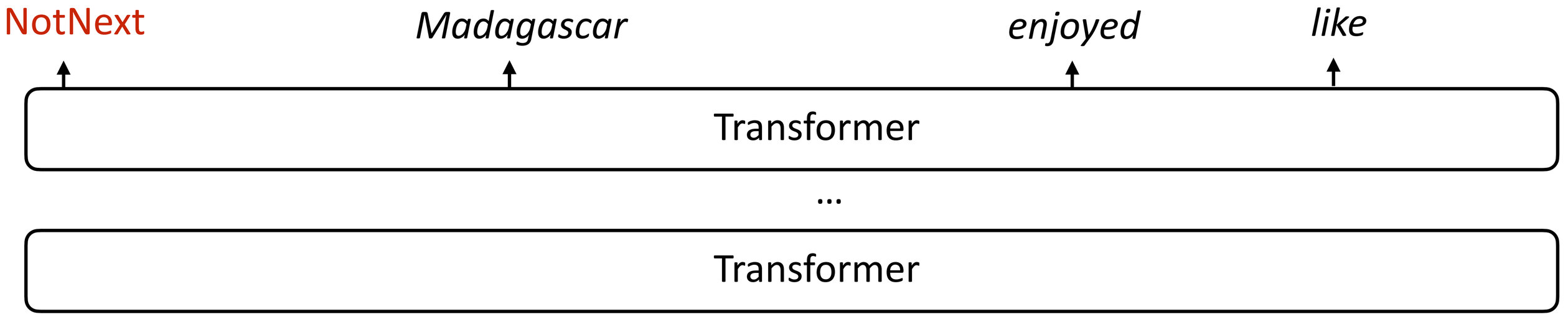
John *visited* **of** *yesterday*

John *visited* Madagascar *yesterday*

Devlin et al. (2019)

Next “Sentence” Prediction

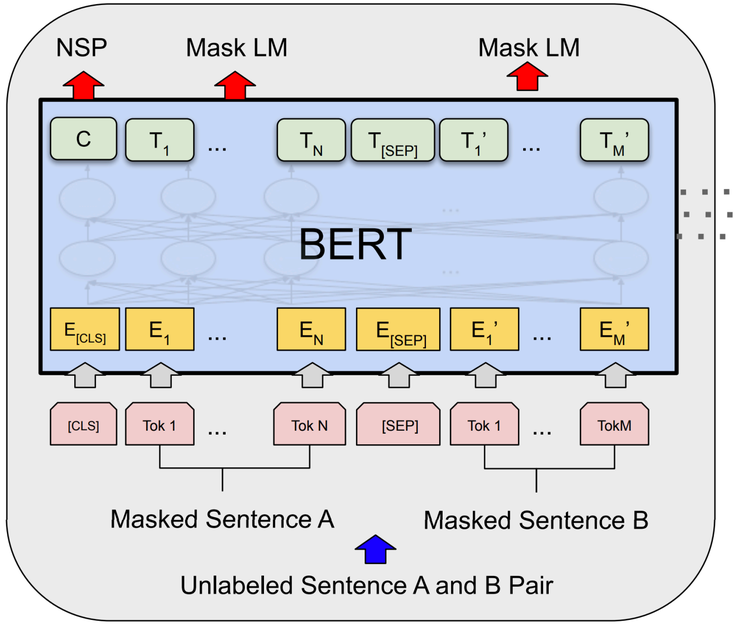
- Input: [CLS] Text chunk 1 [SEP] Text chunk 2
- 50% of the time, take the true next chunk of text, 50% of the time take a random other chunk. Predict whether the next chunk is the “true” next
- BERT objective: masked LM + next sentence prediction

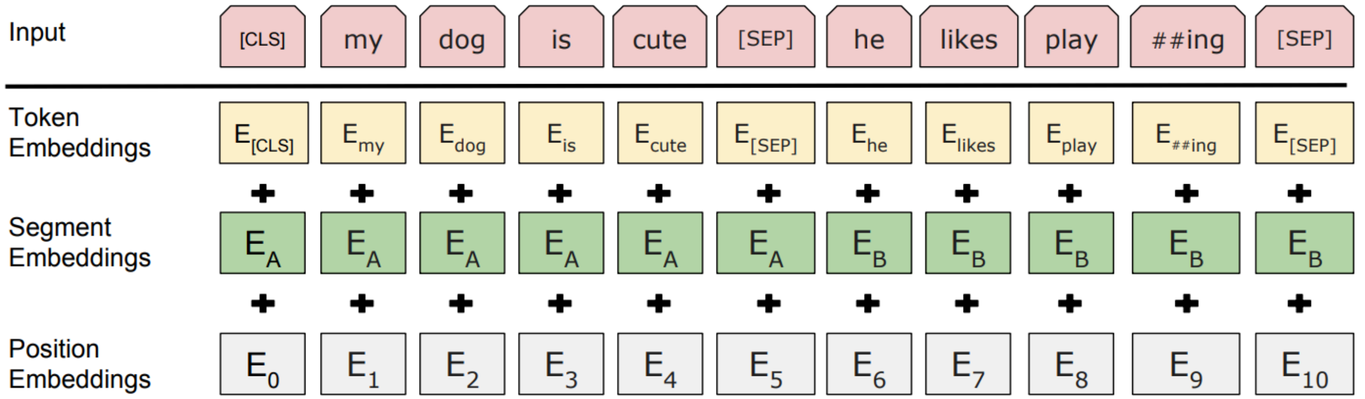


Devlin et al. (2019)

BERT Architecture

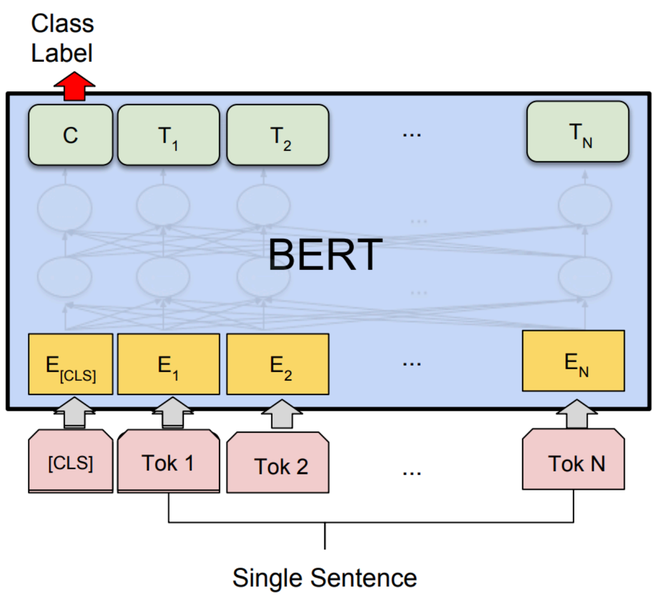
- BERT Base: 12 layers, 768-dim, 12 heads. Total params = 110M
- BERT Large: 24 layers, 1024-dim, 16 heads. Total params = 340M
- Positional embeddings and segment embeddings, 30k word pieces
- This is the model that gets **pre-trained** on a large corpus



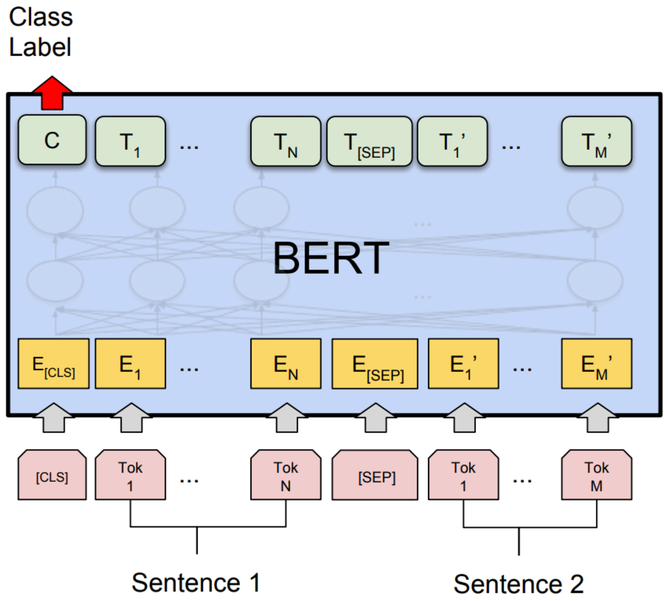


Devlin et al. (2019)

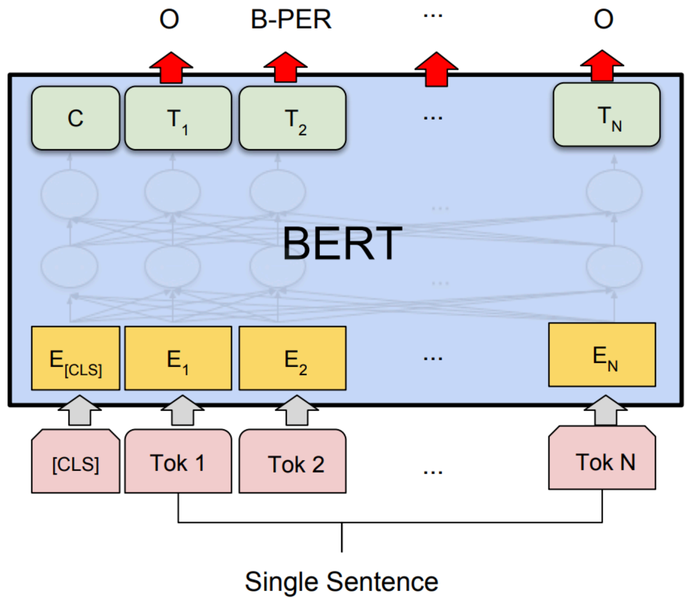
What can BERT do?



(b) Single Sentence Classification Tasks: SST-2, CoLA



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG



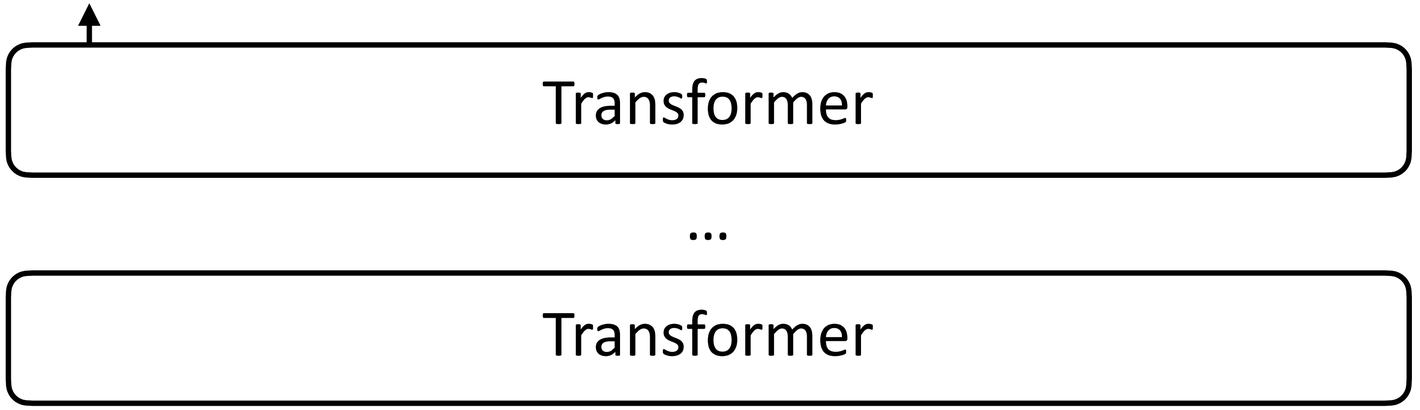
(d) Single Sentence Tagging Tasks: CoNLL-2003 NER

- CLS token is used to provide classification decisions
- Sentence pair tasks (entailment): feed both sentences into BERT
- BERT can also do tagging by predicting tags at each word piece

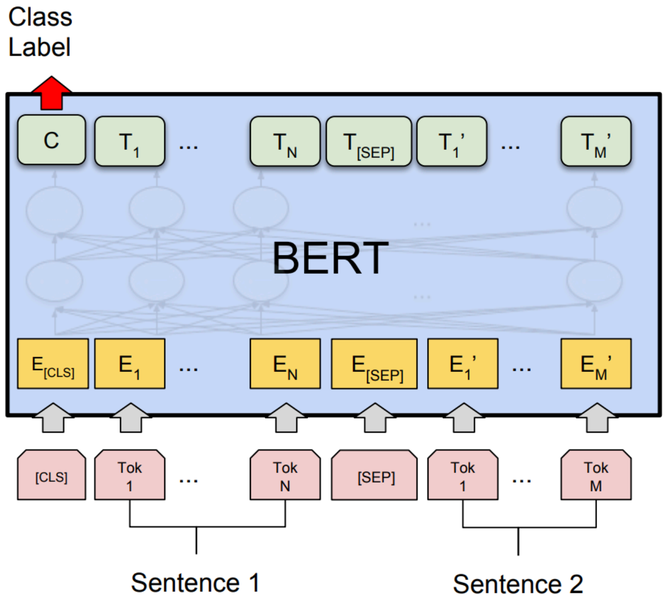
Devlin et al. (2019)

What can BERT do?

Entails (first sentence implies second one is true)



[CLS] A boy plays in the snow [SEP] A boy is outside



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

- How does BERT model this sentence pair stuff?
- Transformers can capture interactions between the two sentences, (even though the NSP objective doesn't really cause this to happen).

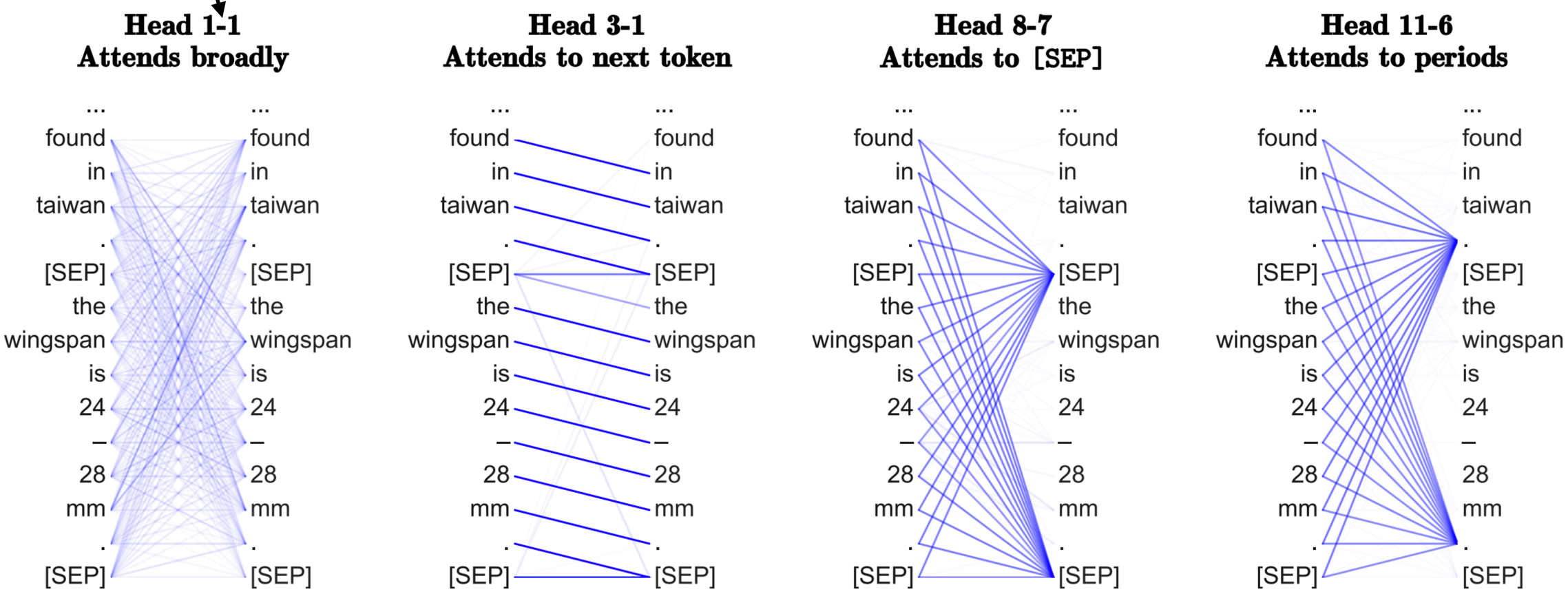
Devlin et al. (2019)

What can BERT NOT do?

- BERT **cannot** generate text (at least not in an obvious way)
  - Can fill in [MASK] tokens, but can't generate left-to-right (you can put [MASK] at the end, then predict repeatedly, but this is slow)
- Masked language models are intended to be used primarily for "analysis" tasks, e.g., sequential tagging, semantic similarity between two sentences, …

What does BERT learn?

layer-head



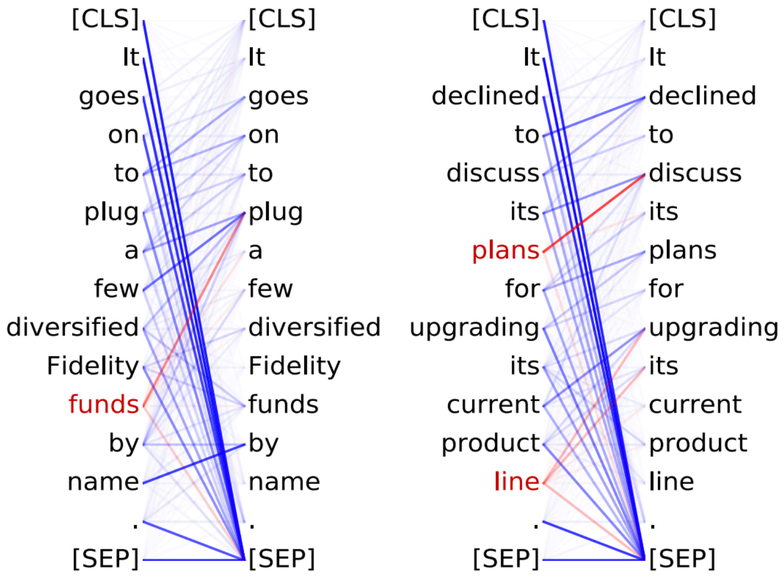
- Heads on transformers learn interesting and diverse things: content heads (attend based on content), positional heads (based on position), etc.

Clark et al. (2019)

What does BERT learn?

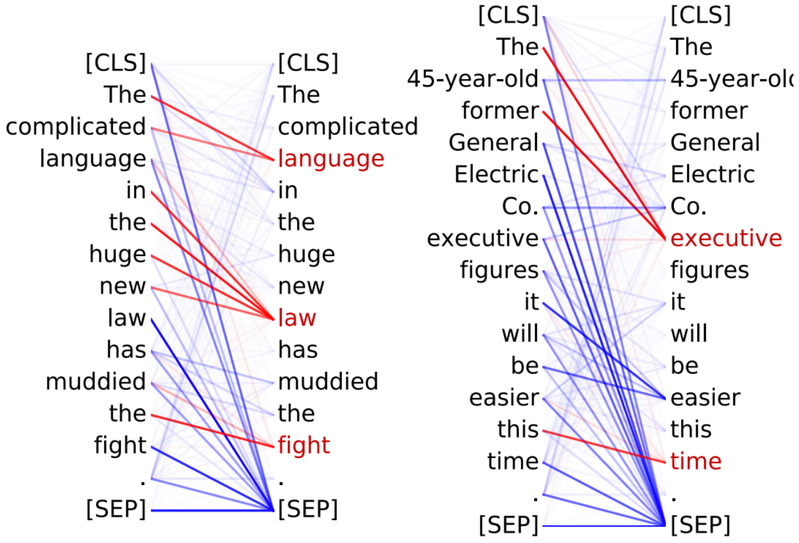
Head 8-10

- **Direct objects** attend to their verbs
- 86.8% accuracy at the `dobj` relation



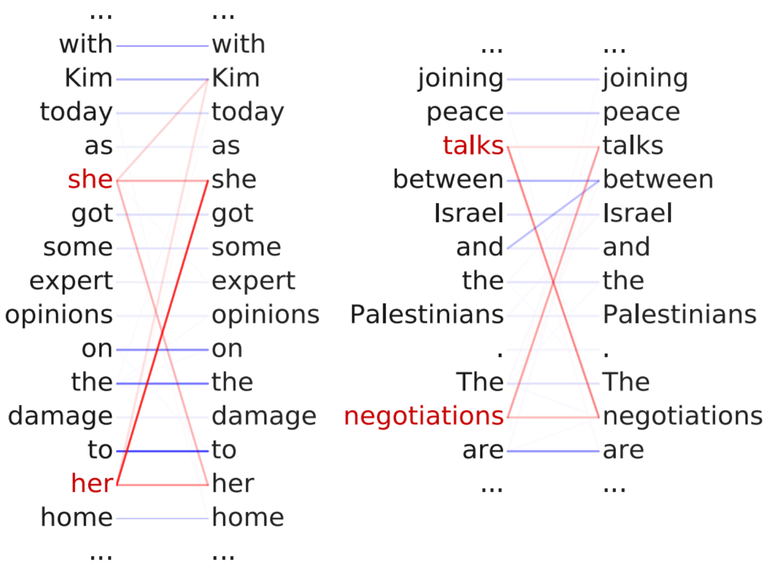
Head 8-11

- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the `det` relation



Head 5-4

- **Coreferent** mentions attend to their antecedents
- 65.1% accuracy at linking the head of a coreferent mention to the head of an antecedent



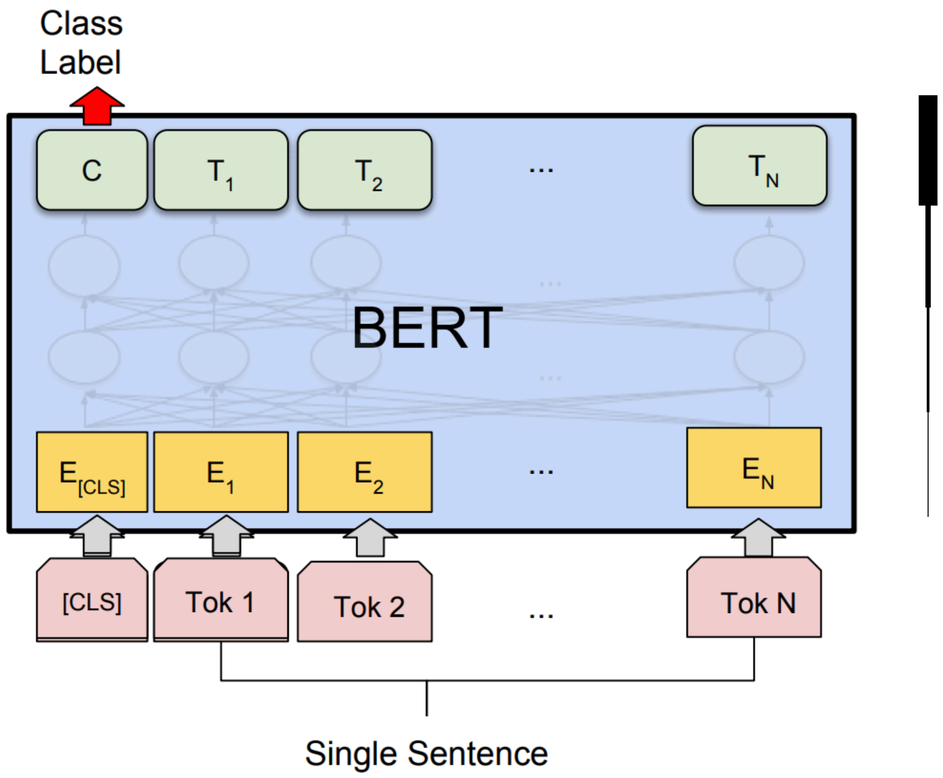
- Still way worse than what supervised systems can do, but interesting that this is learned organically

Clark et al. (2019)

BERT Results, Extensions

Fine-tuning BERT

- Fine-tune for 1-3 epochs, small learning rate (e.g. 2e-5 - 5e-5)



(b) Single Sentence Classification Tasks: SST-2, CoLA

- Large changes to weights up here (particularly in last layer to route the right information to [CLS])
- Smaller changes to weights lower down in the transformer
- Small LR and short fine-tuning schedule mean weights don't change much
- More complex "triangular learning rate" schemes exist

Pre-training, then Fine-tuning

$$\theta_{\text{init}} \xrightarrow{\text{Pre-training}} \theta_{\text{LM}} \xrightarrow{\text{Fine-tuning}} \theta_{\text{FT}}$$

Random initialization | Pre-trained LM | Fine-tuned model

- Masked Language Modeling (e.g., BERT):

$$\mathcal{L}_{\text{MLM}}(\theta) = -\sum_{t \in M} \log p_\theta(x_t \mid \tilde{\mathbf{x}})$$

corrupted input sequence (e.g., with [MASK] tokens)

- Fine-tuning: Learn how to perform a task from labeled examples.

$$\mathcal{L}_{\text{finetune}}(\theta) = -\sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}_{\text{task}}} \log p_\theta(\mathbf{y} \mid \mathbf{x})$$

initialize from θ_{LM}

Tasks: sentence classification, NER, etc.

Evaluation: GLUE

Corpus	|Train|	|Test|	Task	Metrics	Domain
			Single-Sentence Tasks		
CoLA	8.5k	**1k**	acceptability	Matthews corr.	misc.
SST-2	67k	1.8k	sentiment	acc.	movie reviews
			Similarity and Paraphrase Tasks		
MRPC	3.7k	1.7k	paraphrase	acc./F1	news
STS-B	7k	1.4k	sentence similarity	Pearson/Spearman corr.	misc.
QQP	364k	**391k**	paraphrase	acc./F1	social QA questions
			Inference Tasks		
MNLI	393k	**20k**	NLI	matched acc./mismatched acc.	misc.
QNLI	105k	5.4k	QA/NLI	acc.	Wikipedia
RTE	2.5k	3k	NLI	acc.	news, Wikipedia
WNLI	634	**146**	coreference/NLI	acc.	fiction books

Wang et al. (2019)

Fine-tuning BERT

- How does frozen (❄️) vs. fine-tuned (🔥) compare?

Pretraining	Adaptation	NER CoNLL 2003	SA SST-2	Nat. lang. inference MNLI	Nat. lang. inference SICK-E	Semantic textual similarity SICK-R	Semantic textual similarity MRPC	Semantic textual similarity STS-B
Skip-thoughts	❄️	-	81.8	62.9	-	86.6	75.8	71.8
ELMo	❄️	91.7	**91.8**	**79.6**	**86.3**	**86.1**	**76.0**	**75.9**
	🔥	**91.9**	91.2	76.4	83.3	83.3	74.7	75.5
	Δ=🔥-❄️	0.2	-0.6	-3.2	-3.3	-2.8	-1.3	-0.4
BERT-base	❄️	92.2	93.0	**84.6**	84.8	86.4	78.1	82.9
	🔥	**92.4**	**93.5**	**84.6**	**85.8**	**88.7**	**84.8**	**87.1**
	Δ=🔥-❄️	0.2	0.5	0.0	1.0	2.3	6.7	4.2

- BERT is typically better if the whole network is fine-tuned, unlike ELMo

Peters, Ruder, Smith (2019)

Results

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	**Average** -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.9	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	88.1	91.3	45.4	80.0	82.3	56.0	75.2
$\text{BERT}_{\text{BASE}}$	84.6/83.4	71.2	90.1	93.5	52.1	85.8	88.9	66.4	79.6
$\text{BERT}_{\text{LARGE}}$	**86.7/85.9**	**72.1**	**91.1**	**94.9**	**60.5**	**86.5**	**89.3**	**70.1**	**81.9**

- Huge improvements over prior work (even compared to ELMo)
- Effective at "sentence pair" tasks: textual entailment (does sentence A imply sentence B), paraphrase detection

Devlin et al. (2018)

Subsequent Improvements to BERT

- Dynamic masking: standard BERT uses the same MASK scheme for every epoch, RoBERTa recomputes them

  epoch 2

  epoch 1

  … *John* *visited* *Madagascar* *yesterday* …

- Whole word masking: don't mask out parts of words (word pieces)

  … *_John* *_visited* *_Mada* *gas* *car* *yesterday* …

Liu et al. (2019)

RoBERTa

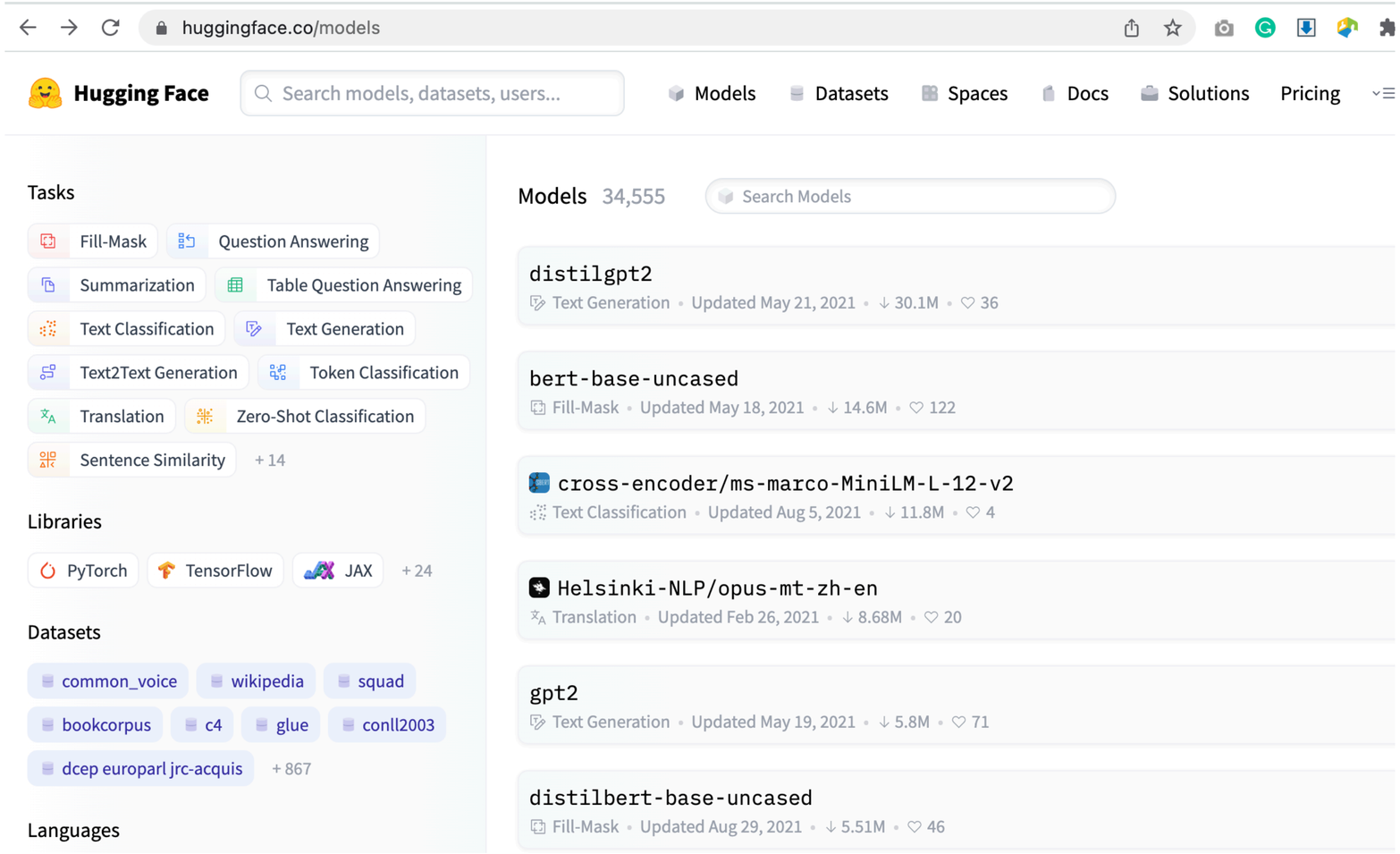
- "Robustly optimized BERT" incorporating some of these tricks
- 160GB of data instead of 16 GB
- New training + more data = better performance

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	**94.6/89.4**	**90.2**	**96.4**
$\text{BERT}_{\text{LARGE}}$						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7

Liu et al. (2019)

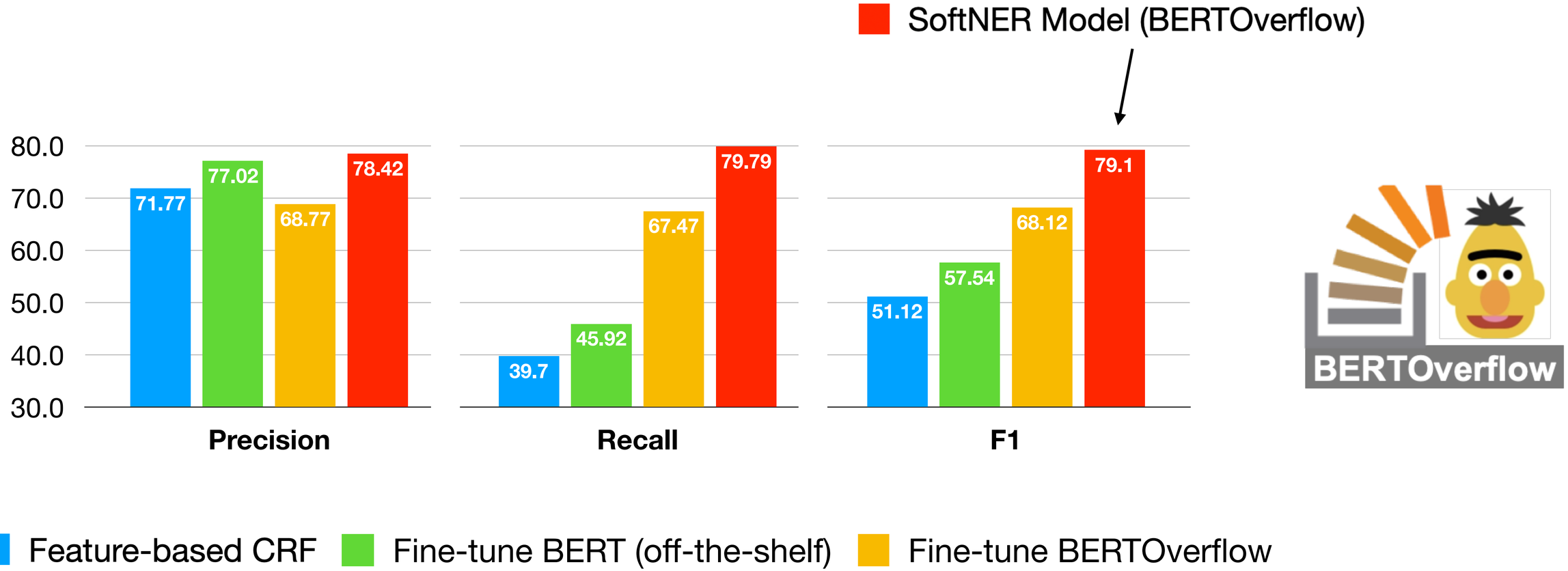
many BERT variations

- For specific text domains (e.g. StackOverflow), or specific languages



BERTOverflow

- A domain-specific BERT model that pre-trained on 10-year StackOverflow data (152M sentences; ~2B tokens).



Jeniya Tabassum, Mounica Maddela, Alan Ritter, Wei Xu. “Code and Named Entity Recognition in StackOverflow” (ACL 2020)

NER in StackOverflow

I am passing an Library_Class: array list as message header to Library_Class: camel route

through Language: java Library_Class: bean as follows

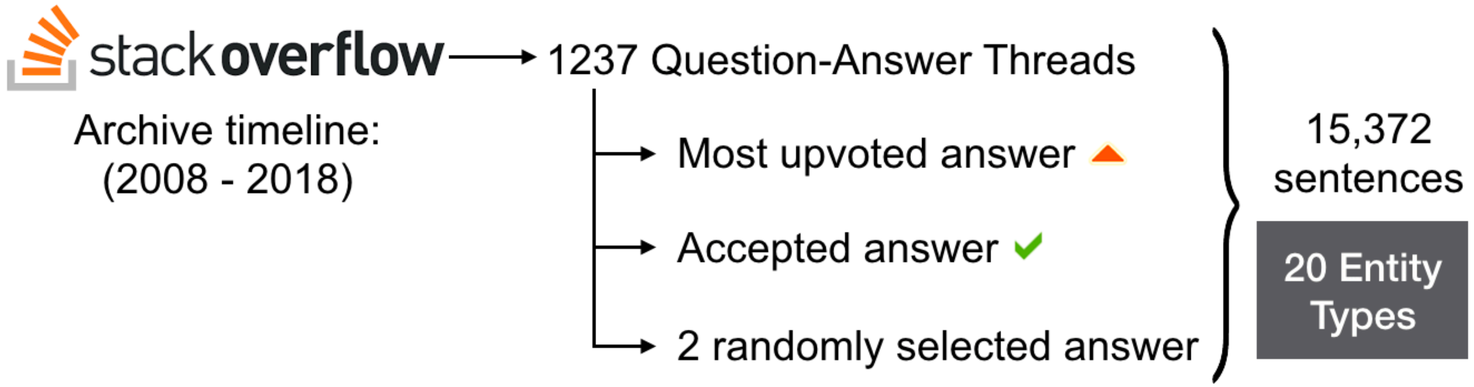
```
ArrayList<String> list=new ArrayList<String>();
                list.add("http://www.google.com");
                list.add("http://www.stackoverflow.com");
                list.add("http://www.tutorialspoint.com");
                list.add("http://localhost:8080/sampleExample/query");
                  exchange.getOut().setHeader("endpoints",list);
```

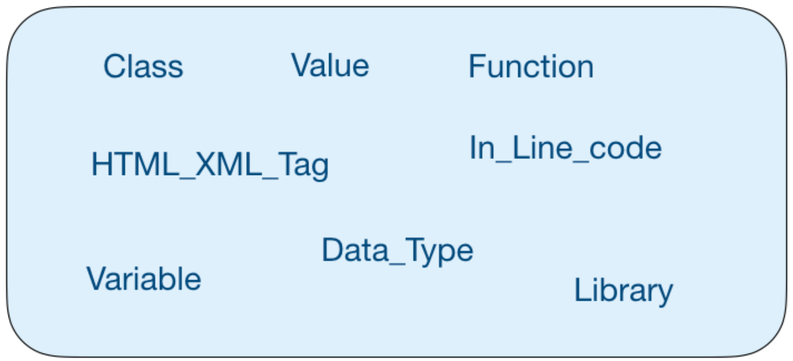
and, inside Library_Class: camel route i want to iterate through this Variable_Name: list

Jeniya Tabassum, Mounica Maddela, Alan Ritter, Wei Xu. “Code and Named Entity Recognition in StackOverflow” (ACL 2020)

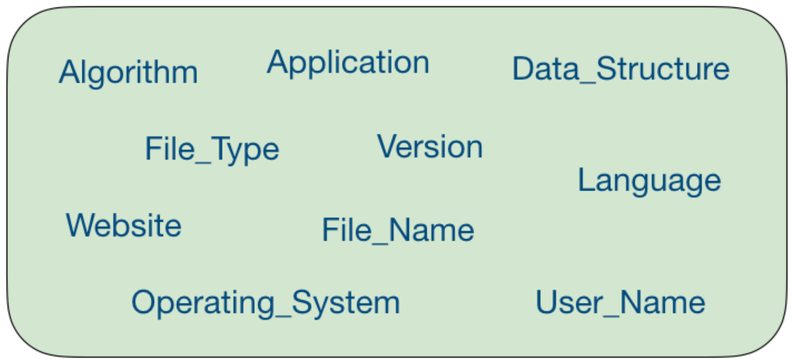
NER in StackOverflow

StackOverflow NER Corpus





Code Entity Types



Natural Language Entity

Jeniya Tabassum, Mounica Maddela, Alan Ritter, Wei Xu. “Code and Named Entity Recognition in StackOverflow” (ACL 2020)

Multilingual BERT

Bilingual BERT

- A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning

	Data Source	**Data Size** (All / English / Arabic)	**IE Performance** (F1 score)
AraBERT (AUBeirut 2019)	News	2.5B / 0B / 2.5B	97.1 / —
mBERT (Google 2018)	Wiki	21.9B / 2.5B / 0.15B	75.3 / 30.1
XLM-RoBERTa (Facebook 2019)	Common Crawl	295B / 55.6B / 2.9B	79.2 / 40.4
GigaBERT (our work)	News, Wiki, Common Crawl	10.4B / 6.1B / 4.3B	84.3 / 48.2

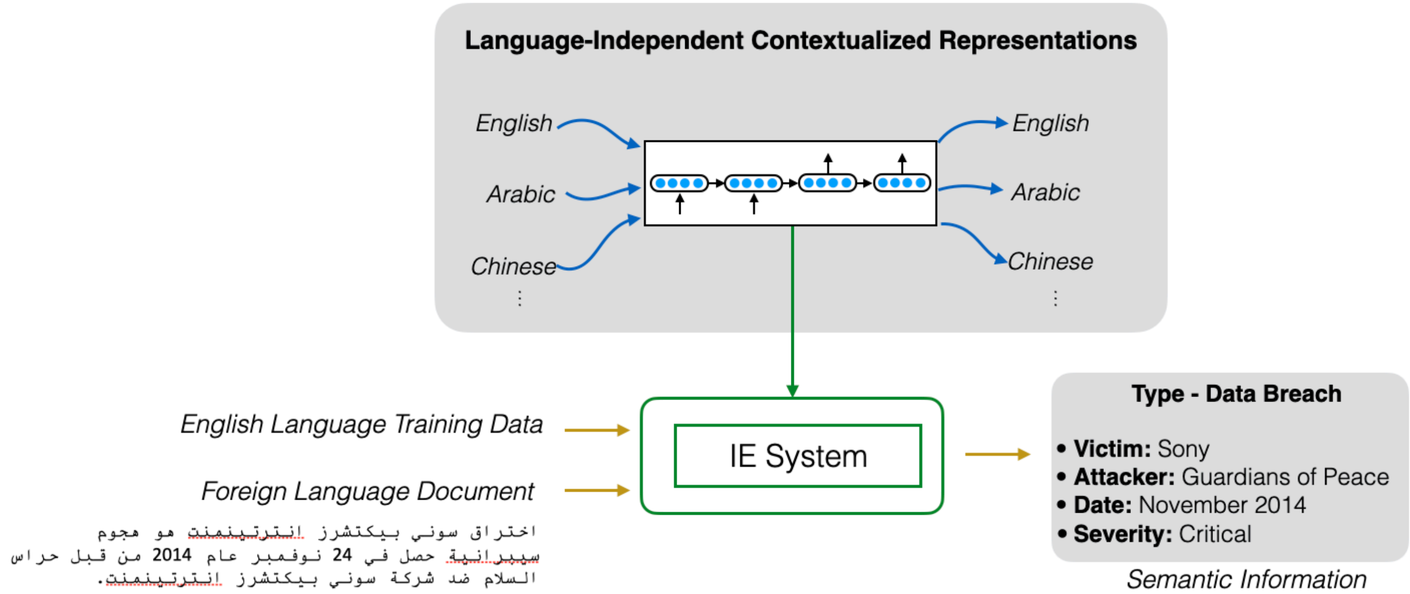
supervised learning

zero-shot transfer learning

Lan et al. (2020)

Information Extraction

- A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning
- i.e., Information extraction models, trained on annotated English data, directly apply to non-English texts to extract entities and events.



IARPA

Lan et al. (2020)

Cross-lingual Transfer Learning

- Pre-train LMs on unlabeled multilingual data, then fine-tune for specific tasks using labeled data in some languages (e.g., English)

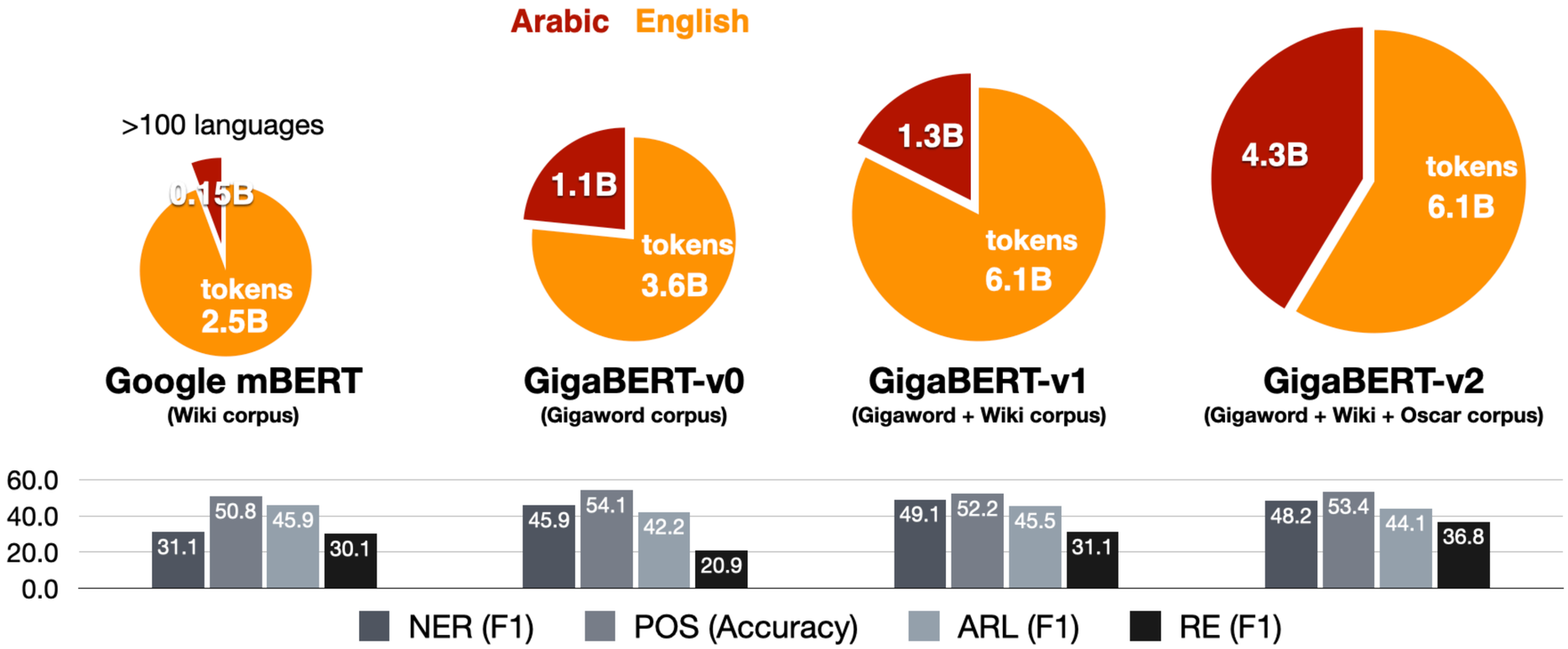
$$\theta_{\text{init}} \xrightarrow[\text{Unlabeled multilingual data}]{\text{Pre-training}} \theta_{\text{mLM}} \xrightarrow[\text{Labeled source-language data}]{\text{Fine-tuning}} \theta_{\text{FT}}^{\text{src}}$$

- Apply the fine-tuned model directly to target languages (e.g., Arabic), without labeled data in target languages or further training:

$$\hat{\mathbf{y}}_{\text{tgt}} = \arg\max_{\mathbf{y}} p_{\theta_{\text{FT}}^{\text{src}}}(\mathbf{y} \mid \mathbf{x}_{\text{tgt}})$$

Bilingual BERT

- A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning



Lan et al. (2020)

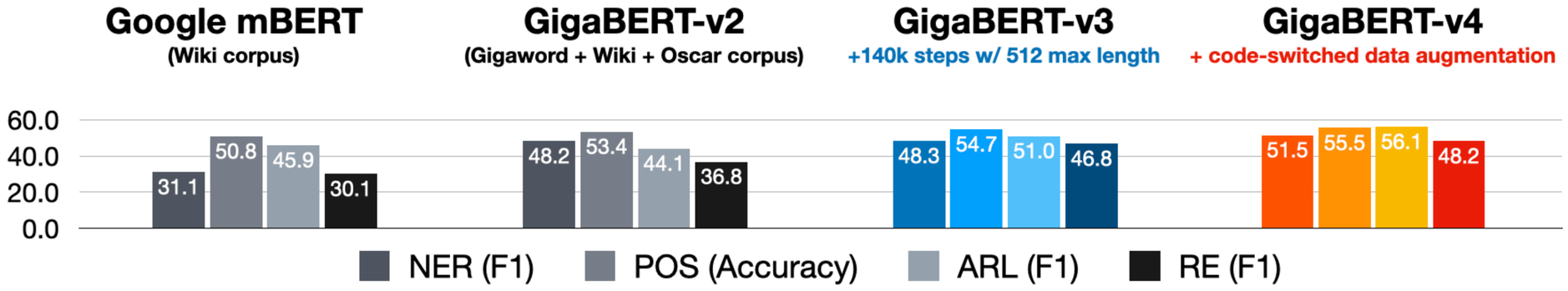
Bilingual BERT

- A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning + synthetic code-switching data

(English Translation — Official: China reaches the five-year plan target of cutting emissions)

مسؤول: **الصين** تبلغ هدف **الخطة** الخمسية المتعلق بخفض **الانبعاثات**

مسؤول: China تبلغ هدف plan الخمسية المتعلق بخفض emissions



Lan et al. (2020)

mmBERT

- Multi-stage training with different number of languages per stage

		Pre-training		Mid-training		Decay Phase	
Category	**Dataset**	**Tokens (B)**	**%**	**Tokens (B)**	**%**	**Tokens (B)**	**%**
Code	Code (ProLong)	–	–	–	–	2.8	2.7
Code	Starcoder	100.6	5.1	17.2	2.9	0.5	0.5
Crawl	DCLM	600.0	30.2	10.0	1.7	–	–
Crawl	DCLM (Dolmino)	–	–	40.0	6.7	2.0	2.0
Crawl	FineWeb2	1196.6	60.2	506.7	84.3	78.5	76.0
Instruction	Tulu Flan	15.3	0.8	3.1	0.5	1.0	1.0
Math	Dolmino Math	11.2	0.6	4.3	0.7	0.5	0.5
Reference	Books	4.3	0.2	3.9	0.7	2.2	2.1
Reference	Textbooks (ProLong)	–	–	–	–	3.1	3.0
Reference	Wikipedia (MegaWika)	4.7	0.2	1.2	0.2	9.5	9.2
Scientific	Arxiv	27.8	1.4	5.4	0.9	3.3	3.2
Scientific	PeS2o	8.4	0.4	3.2	0.5	–	–
Social	StackExchange	18.6	0.9	3.0	0.5	–	–
Social	StackExchange (Dolmino)	1.4	0.1	2.8	0.5	–	–
Total		1989.0	100.0	600.8	100.0	103.3	100.0

Marone et al. (2025)

mmBERT

- Pre-trained on 3T tokens of multilingual text in over 1800 languages

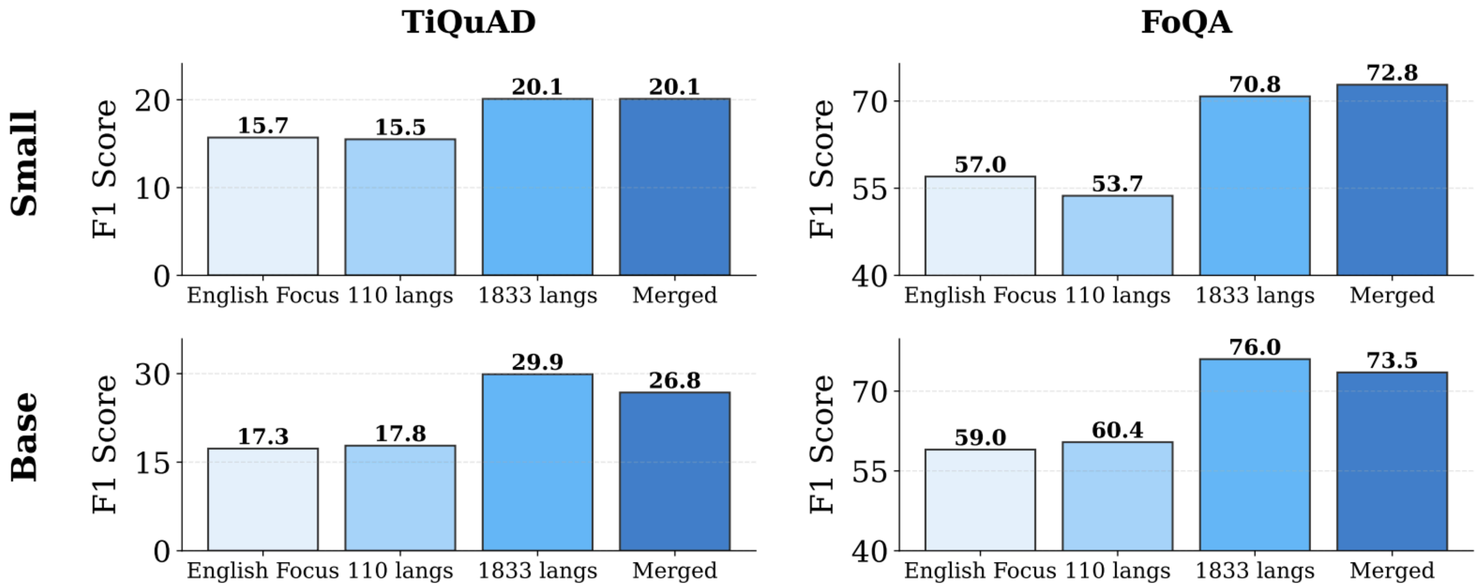


Figure 2: Performance of models using different decay phases on two languages (Tigray and Faroese) only added during the decay phase. We see that MMBERT with the 1833 language decay phase shows rapid performance improvements despite only having the models in the last 100B tokens of training. The final MMBERT models shows improvements by merging together checkpoints.

Marone et al. (2025)

Other Encoder-only LMs

- XLM-RoBERTa (Conneau et al., 2020) for 100 languages
- mDeBERTa (He et al., 2021) for 100 languages
- MosaicBERT (Portes et al., 2023; Nussbaum et al., 2024)
- ModernBERT (Warner et al., 2024)
- EuroBERT (Boizard et al., 2025) for 15 languages
- NeoBERT (Le Breton et al., 2025)
- mmBERT (Marone et al., 2025) for 1800 languages
- ? Jev (TypeSafe AI, 2026)